# Loki III: Reverse Engineering Structural Netlists without a Language Model
# How far do provenance, a Bayesian L1 and a graph L2 get on their own?

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Abstract—Gate-Breaker showed that a language model can recover functional RTL from a gate-level netlist, and it ended by naming the graph techniques its model needed: clustering to name buses, a classification of signals as datapath or control, and partitioning. Loki III asks the complementary question: how much structure can be recovered without a language model, from provenance, a Bayesian hypothesis flow and graph analysis alone? This paper describes the benchmarks built to answer that question and reports the recovery statistics under the Gate-Breaker metrics. The benchmarks are 68 synthesizable micro designs and the 156 VerilogEval references (150 build on three floors), compiled by a provenance-tracked front end into named, labelled, anonymised netlist datasets; a renamed-instance protocol then proves the pipeline naming-neutral across 105,646 renamed identifiers. On the L1 hypothesis flow, the structures that are already reverse engineered by the synthesis flow recover almost perfectly (hard adders 67 of 67, hard multipliers 30 of 30), while the structures the flow erased recover weakly (macro-F1 0.354 on the first floor, 0.108 on the second, calibrated). A second round of compute-only experiments finds that re-mapping the gate netlist to LUT6 brings its in-sample recovery to the second floor's level (not on the held-out designs, where the result is inconclusive), that the language model is more readable than Loki on the same designs while Loki's output is formally equal to its input, and that an operator-recovery F1 sees what AST and CFG similarity cannot. A third round, also compute-only and with its decision rules fixed beforehand, finds that most wrong candidates of the uncalibrated detectors lie inside real constructs, but only by the letter of the rule: the result does not survive leaving out one design or the near-tautological enable fragments, and the fragments rarely tile the words. The L2 graph stage beats a 1-WL baseline on real micros once node, graphlet, WL and NPN features combine (macro-F1 0.4241 against 0.2717) and its clusters carry the designer's hierarchy (Louvain modularity 0.4220). We state the honest limits: no language model step exists, the recovered structure is structural rather than functional, and the paper's own determinism defects are reported.

Index Terms—reverse engineering, netlist, provenance, Bayesian inference, graph neural networks, FPGA CAD

## I. Introduction

Gate-Breaker [1] takes a gate-level netlist with obfuscated wire names, formats it into a four-step prompt (interface, decomposition, function, generation), asks a language model to write RTL, and checks the result by simulation. On 156 VerilogEval problems [2] its best model recovered functional RTL for 75.1% of the runs, and 53.4% on the hardest quartile. The paper's own conclusion is the interesting part: external graph tools should help the model – clustering to name buses, a classification of signals as datapath or control, and partitioning of the netlist into smaller parts.

Loki III (https://github.com/drpaj12/loki) is the Yggdrasil III suite's reverse-engineering tool, and it targets exactly those three items – with no language model step at all. Its pipeline is L0, a provenance ground-truth dataset made from the benchmarks; L1, a Bayesian hypothesis flow whose pattern detectors propose structures and whose proofs confirm them; and L2, graph analysis (clustering, structural features, graph neural networks) over the same netlists. The research question this paper answers is: how good are the L0, L1 and L2 experiments, without more intelligence, at recovering structure? The answer matters for two audiences. For the Gate-Breaker line, it prices the graph side of the future work the paper names. For reverse engineering, it measures what a provenance-tracked front end and honest statistical machinery can do before any model is added.

The contributions are: (1) a benchmark method – provenance-tracked, labelled, anonymised netlist datasets with a renamed-instance protocol that proves naming neutrality, so the pipeline can be evaluated on designs whose names carry nothing; (2) the L1 and L2 recovery statistics under the Gate-Breaker metrics and under Loki's own (precision, recall, calibration, clustering quality, GNN against a 1-WL baseline); (3) the honest limits, including two determinism defects the experiments found.

## II. Background

### A. The provenance problem and Odin III

Synthesis modifies the designer's functional intention: flattening removes hierarchy, optimization merges oper-

ators, and mapping covers logic with cells that carry no source names. Linking each netlist object back to the HDL that produced it is the provenance problem [3]. Odin III, the provenance-tracked front end under Loki, keeps that link: every object of its one IR records its source location, and the link survives elaboration, optimization, partial mapping and the linked ABC flow. Loki stands on that link: its L0 datasets carry the answer key the synthesis tool kept, which is what makes the recovery statistics in this paper possible at all.

### B. Gate-Breaker as the method baseline

Gate-Breaker measures recovery by functional correctness and by the similarity of the recovered and reference RTL: Success Rate (the fraction of $K$ samples that simulate correctly), Success@$k$ (the fraction of designs solved within the first $k$ samples), AST similarity (mean subtree Jaccard over normalized identifiers, depth $d = 3$) and CFG similarity (a normalized shortest-path node kernel over control-flow graphs). Its complexity score weights sequential blocks, generates, loops, instantiations, always blocks, buses, multiplexers, cases, arithmetic operators, assignments and concatenations into quartiles Q1 (simple combinational) through Q4 (complex sequential). Loki reports its own statistics on the same axes where they apply – recovery rate per structure kind, calibration error, clustering quality against the known hierarchy, and GNN accuracy against a 1-WL baseline – and the Gate-Breaker metrics on the replication runs.

## III. Methodology: floors, pipeline and evaluation

### A. Floors: the abstraction level of the netlist

A floor is the abstraction level and technology mapping at which an Odin III design is cut and written out as the L0 netlist, the object the reconstruction receives. Every floor starts from the same elaborated design, the common front proc; fsm; opt; flatten; opt -full; wreduce (FRONT in loki/odin.py), and then maps it with a floor-specific script (floor_script in the same file). Table I lists the floors. Four exist: f1-k6 and f1-earch (one floor per FPGA architecture file; the directory name is f1- plus the architecture's lowercase prefix), f2, and f3.

What a floor's netlist contains decides what is left to reconstruct. At f1 the flow's hardmap step has already recognised adders, multipliers and memories, so they appear as typed hard cells; the construct they implement is part of the netlist, and the labels still name it. At f2 the same logic is only LUT truth tables: an adder is a carry chain of LUTs that has to be found again. At f3 the logic is a network of two-input gates, the form Gate-Breaker's language model receives. The f2 output does not depend on the architecture file, and f3 reads none, so their files are byte-identical across architectures.

Loki uses several floors because what a synthesis flow erases depends on the mapping, and recovery has to be measured against each. f1 keeps the structures the flow already reverse engineered, which is what lets the hard detector of L1 read them from Odin III's unmap facts; f2 removes that crutch with the same source and the same answer key; f3 makes the input comparable to Gate-Breaker's and is the floor on which the AST and CFG similarities are computed (Section V-C). Within one design the floors share one source of truth, the provenance spans of the same elaborated IR, so a difference between floors is a difference of the mapping.

### B. L0: provenance ground-truth datasets

The first benchmark set is the 68-design micro suite (counters, registers, arithmetic, ALUs, muxes, comparators, small FSMs) that Odin III already compiles and checks against Yosys; the second, the 156 VerilogEval references, is described in Section IV. L0 compiles each design on every floor of Table I. Every netlist object carries its provenance (source file and line), a neutral n$k$ name from one shared counter, and a label: the structure kind (construct_type) is the answer key in labels.csv, and the netlist-visible hard-block kind (cell_type) travels in the graph labels, so structures that are already reverse engineered stay visible to the reconstruction. loki check validates every dataset: 68 designs, 269 floors, a label row for every node, zero errors. The two hard-block designs the v1.0 front end refused now compile on the v1.1.0 SystemVerilog front end, so the set is complete.

Renamed instances. A separate protocol asks the naming question directly: rename every identifier of every netlist (105,646 identifiers across 269 netlists) through a shared permutation, run the pipeline on the renamed floors, and inverse-map every token back. The verifier reports token-for-token equality with the original, zero failures, and the base and renamed score digests are byte-identical (sha256 over all run scores, 204 of 204 pairs) – the pipeline is naming-neutral, so nothing it recovers is an artefact of a name.

### C. L1: the Bayesian hypothesis flow

L1 is a round-based hypothesis flow over one netlist, with no language model. It reads a floor netlist into a fresh Odin III design (level 0, never edited) in which every cell is a function, and keeps a working copy that grows one level of hierarchy per committing round.

Hypotheses and detectors. A hypothesis is a claim that a set of level-0 cells is a structure of some kind, for example: a flip-flop with enable or synchronous reset (ff_en, ff_srst), a register, a bus of isomorphic slices, an adder, subtractor, comparator or constant comparison (eq_const), a mux or mux_tree, a counter, shift_reg, lfsr or fsm, a datapath or control unit, a per-cell role, or a hard block (hard_add, hard_mul, hard_mem). Nine detector plugins (arith, bus, ff_pattern, fsm, hard, mux, role, seqfn, unit) propose them. A detector reports features as bin labels (a width of 5 to 8, whether an enable function is

TABLE I
The floors of L0. All floors run the common front first; the mapping script is the second part of floor_script. Node kinds are those of graph.json (pi, po, const, lut, ff, hard, gate).

| Floor | Reads | Mapping script | Netlist contents |
|---|---|---|---|
| f1-k6 | read_arch k6_frac_N10_ frac_chain_mem32K_40nm | hardmap; lower; abc –lut 6 | LUT6, flip-flops, and hard adder, multiplier and RAM cells that keep their Odin III type |
| f1-earch | read_arch of the EArch file | same as f1-k6 | same kinds, EArch hard blocks |
| f2 | read_arch (cell types only) | lower; abc –lut 6 | LUT6 and flip-flops only; arithmetic is soft, so no hard node exists |
| f3 | read_techlib lib/cmos6.o3lib | lower; abc –gates cmos6; dfflibmap cmos6 | library gates (BUF, INV, NAND2, NOR2) and DFF flip-flops |

exact, the NPN class of a cell) and never a likelihood; the likelihoods come from calibration.

Bayesian inference. Each hypothesis $h$ of kind $k$ from detector $d$ carries a log-odds $L(h)$, the logarithm of the odds that it is a real structure, and $p(h) = 1/(1+e^{-L(h)})$ is its probability. $L$ is a naive-Bayes sum: a prior plus evidence items, each the log-likelihood ratio (LLR) of an observation,

$$L(h) = \pi_{d,k} + \mathrm{clip}_{\pm 6.91}\Big( \sum_f w_{\phi(f)}\, \lambda_{k,f}(b_f) + \min\big(1.10\, n_{\mathrm{conf}},\, 3.30\big) - 2.30\, n_{\mathrm{ref}} \Big) + e_h, \quad (1)$$

where $\pi_{d,k}$ is the prior log-odds of that detector and kind; $\lambda_{k,f}(b_f)$ is the LLR of feature $f$ in bin $b_f$; $w_{\phi(f)}$ weights the feature's family (structural, WL colour, NPN class or L2; the f3 calibration uses the first three only); $n_{\mathrm{conf}}$ and $n_{\mathrm{ref}}$ are the numbers of confirmed (or committed) and refuted child hypotheses (a bus slice, a register's flip-flops), which raise and lower their parent; and $e_h$ collects fixed records (a sampled check adds $\ln 100 = 4.61$). The clip at $\pm \ln 1000 = 6.91$ limits the double counting that a naive-Bayes sum of correlated features would otherwise allow. A proof sets $L(h) \geq \ln 9999 = 9.21$ (the odds become 9999 to 1), and a counterexample sets $L(h) = -\infty$.

All three tables are calibrated from the training split of L0 (loki calibrate). The flow runs with default tables on every training design; each hypothesis is labelled true when it matches an L0 construct (Section III-E). The prior is the Laplace-smoothed precision of the detector, $\pi_{d,k} = \mathrm{logit}\,\frac{tp+1}{n+2}$ over $n$ proposals of which $tp$ are true. For example, the f3 comparator detector made 31,036 proposals of which 2 were true, so $\pi = \ln\frac{3}{31035} = -9.24$; the committed value is $-9.244259$ (experiments/f3-recovery/calib/f3.json). A bin's LLR is $\lambda = \ln\frac{(pos+1)/(POS+B)}{(neg+1)/(NEG+B)}$, with $pos$ and $neg$ the true and false proposals in the bin, $POS$ and $NEG$ the totals of the kind, and $B$ the number of bins. The family weights $w$ are the grid values in $\{0, 0.1, \ldots, 1\}$ that maximise the Bernoulli log-likelihood of the labels, fitted by three sweeps of coordinate ascent on out-of-fold rows (two folds, split by design). Without calibration (–calib none) the priors are 0 and the LLRs are the defaults of bayes.py. The reliability check is the expected calibration error over ten equal-width bins of the predicted probability, $\mathrm{ECE} = \sum_b \frac{n_b}{N} |\bar{p}_b - r_b|$ with $r_b$ the observed match rate.

Proof engines. A hypothesis with a functional claim ("these 9 LUTs compute $A + B$ at width 5") can be proved. Three engines prove claims: cone_check first when the claim has a template, the truth table otherwise or when the cone cannot be bound to the template, and transfer after an exact proof. (i) cone_check, an Odin III pass, proves that the claim's cone of level-0 cells is equivalent to a template word cell (for example $add, $sub, $eq, $mux) with its ports bound to the cone's boundary nets, under a timeout; the reference functions of the claims (claims.py) cover addition, subtraction, the six comparisons, constant comparison, 2:1 and $N$:1 multiplexers, counters, shift registers and LFSRs. (ii) The truth-table prover enumerates the claim's inputs and side nets when together they number at most 16, an exact proof or a counterexample, and otherwise simulates 4096 seeded random vectors, which can refute a claim but only raises it by $\ln 100$. (iii) Proof transfer groups open hypotheses by operation, parameters and the 1-WL colour of their output drivers; when one member is proved exactly, the others are tried by a lockstep cone_match of the two cones, and one transferred member per group is proved for real as a spot check, revoking the transfers of the group if it fails. A sequential claim is proved as a next-state function (state in, next state out); an FSM gets no proof.

Deciding the structure. Hypotheses that claim the same cells compete. Two hypotheses conflict when both kinds are exclusive, their cell sets intersect, and neither is a strict subset of the other (a strict subset is nesting: the smaller becomes a child of the larger). For each connected component $C$ of the conflict graph the competitive probability is

$$p_{\mathrm{comp}}(h) = \frac{e^{L(h)}}{1 + \sum_{j \in C} e^{L(j)}}, \quad (2)$$

where the "1" is the option that none of them is real; a hypothesis without conflicts has $p_{\mathrm{comp}} = p$. Two confirmed hypotheses with equal cells are two true forms of the same logic and count as one member of $C$; the one whose kind ranks higher (a counter above an adder, for example) wins at commit and the other is kept as an alternative. A proof is spent on hypotheses with $p_{\mathrm{comp}} \geq 0.2$, most

Algorithm 1 The L1 round loop (loki/re/loop.py).

$level_0 \leftarrow$ read netlist as functions; $work \leftarrow level_0$
if floor f1 with hard cells then run unmap on *work*
for $r = 1$ to 8 do
    propose: every detector adds hypotheses over the view of *work*
    score: $L(h)$ by Eq. 1; compete: $p_{\text{comp}}$ by Eq. 2
    prove claims with $p_{\text{comp}} \geq 0.2$: cone_check, truth table, transfer
    re-score and compete; reject $p_{\text{comp}} < 0.01$
    select $p_{\text{comp}} \geq 0.9$ (children first); commit by raise and submod
    if nothing committed, nothing new, $|\Delta L| \leq 10^{-6}$ then
        break
    end if
end for
write design.v, overlay.json, report; equiv-check against the input

probable first, at most 200 per round; a hypothesis below 0.01 is rejected, and re-enters when later evidence lifts it. A hypothesis is selected for commit when $p_{\text{comp}} \geq 0.9$, every child that could commit is already committed, and it does not conflict with a committed piece. Selection is greedy in the order ($p_{\text{comp}}$ descending, kind rank, cell count descending, id), and the losers of a conflict become rejected or alternative.

Commit and the final structure. A selected hypothesis is rebuilt on a copy of the working design: a hypothesis is raised to a single word cell only if its kind is one of adder, subtractor, comparator or multiplexer, its claim has a cone_check template, and it is proved by cone_check (directly or by transfer); the cone is re-checked by cone_check unconditionally before the raise. A claim proved only by truth table is wrapped, not raised, and counters, shift registers, LFSRs, FSMs, registers, constant comparisons, multiplexer trees and carry-in adders are never raised. Every selected piece except buses and roles (which only set attributes) is wrapped by submod in a module named for its kind, whose instance is tagged with the hypothesis id, kind, confidence $p_{\text{comp}}$, proof status and the recovered function. If raise or submod fails, the offending hypothesis is rejected with a record of the error and the commit is retried once without it (reject_failed). The structure that L1 reports is therefore exactly the set of committed pieces in design.v; the alternatives, rejected and open hypotheses stay in overlay.json with their evidence. Algorithm 1 summarises the loop. It stops when a round commits nothing, proposes nothing new and changes no log-odds by more than $10^{-6}$ (a fixed point), after eight rounds, or at the time budget, then flattens and lowers the final design and checks it against the input with equiv-check; a result of differs is a bug and the run exits with an error.

The hard detector reads the unmap pass's facts: a hard adder, multiplier or memory that the flow already reverse engineered is recovered directly, with the decline reasons the pass recorded (carry-in chains and the like) kept as open hypotheses. The hard detector does not run on f2 or f3.

## D. L2: graph analysis

L2 reads the L0 graph.json files and nothing else: never key.json and never the BLIF. It has three parts.

Structural features. Each node gets numeric features (fan-in, fan-out, logic level, LUT width, hard-port count), its kind and, on hard cells, its cell_type, plus three families: colours of 1-WL colour refinement (a hash of the node's label and the multiset of its neighbours' colours, per round, so that equal colours mean equal unfolded neighbourhoods), the NPN class of a LUT's function (invariant under permutation and negation of the inputs and the output), and the ORCA orbit counts of the 4-node graphlets around the node. Before graphlets are counted, the global signals (clock, reset, constants, global enables) are classified by role and their edges removed, so that a clock does not connect everything.

Clustering. Four methods partition a netlist graph into candidate blocks: k-means on the feature space ($k = \text{clip}(\text{round}\sqrt{N/2}, 2, 200)$), affinity propagation on a sample of at most 3000 nodes with the remaining nodes assigned to the nearest exemplar, spectral clustering (a spectral embedding of the largest connected component, then k-means), and Louvain community detection (resolution 1, no $k$). The first two use node features; the last two use the graph structure. A partition is scored by its modularity on the undirected simple projection of the graph, by its NMI against the known module_path and construct_type labels and by its ARI against module_path only over the LUT, flip-flop, hard and gate nodes. These two numbers are a quality report, never an input to a method: modularity says whether the clusters are dense subgraphs, NMI whether they are the designer's modules.

GNN against 1-WL. A node classifier predicts each node's construct_type: a two-layer GraphSAGE (GCN as the variant) of hidden width 64 with ReLU and dropout 0.2 over the graph and its reverse edges, trained with Adam (learning rate 0.01, weight decay $5 \cdot 10^{-4}$) for a fixed 100 epochs, full-batch, with class weights $1/\sqrt{\text{count}}$. The split is L0's, by design, with the class vocabulary taken from the training designs only (rarer values, and values first seen in a test design, are one class ___other___); there is no validation set and no early stopping, so nothing is tuned on a test design. Loss and metrics cover only the LUT, flip-flop, hard and gate nodes with a known target. The question is whether message passing earns anything over what the colour refinement already knows, because a message-passing GNN is at most as discriminating as 1-WL [4]. So the 1-WL baseline predicts, for each node, the most common training class of its WL colour (backing off from round 4 to round 0 when a colour has fewer than five training nodes), and the GNN has beaten it only if its test macro-F1 is higher. Two further baselines are reported on

the same nodes, the majority class and a logistic regression on the features alone. The task is that of GNN-RE [5], which maps each gate of a flat netlist to the module it belongs to; Loki's model is a baseline for the same task, not a re-implementation, and the numbers are not compared with that paper's.

### E. Evaluation protocol: how L1 and L2 are scored

Gate-Breaker evaluates a stochastic model: for each design the model is sampled $K$ times, each sample's RTL is extracted and simulated against the VerilogEval testbench, and the results are aggregated per complexity quartile as the success rate (the fraction of the $K$ samples that pass), Success@$k$ (the fraction of designs with a passing sample among the first $k$), and the AST and CFG similarity of the recovered RTL to the reference. Loki's L1 and L2 are deterministic and their output is a structure, not an RTL text, so the same four questions are asked in different terms. Table II sets the two side by side.

L1, step by step.

1) Split. L0 assigns each design to train or test by a hash of the design name (sha256(seed:design) mod $100 < 80$ is train), so the split is fixed by the name and independent of the run.
2) Calibrate on the training designs of one floor (loki calibrate): the priors, bin LLRs, family weights and reliability curve of Section III-C.
3) Run loki re on every test netlist, calibrated and, as a diagnostic, with –calib none. The run never reads the answer key.
4) Ground truth. From words.json, the L0 word cells give ground-truth instances $(\text{type}, F_g)$ with $F_g$ the set of final cells of the word; counters, mux trees and FSMs are derived from the word connectivity.
5) Count and match. A hypothesis counts when it is committed or its $p_{\text{comp}} \geq 0.5$ (–score-p); it matches a ground-truth instance of an accepted type when the Jaccard of their cell sets is at least 0.5, greedily one-to-one by Jaccard. Precision is matched over counted, recall matched over ground truth, and average precision sweeps the threshold from 0.1 to 0.9.
6) Buses, roles and hierarchy. A bus matches a ground-truth bus (nodes sharing a module path and bus name, at least two bits) at the same Jaccard; its bit order is scored exactly and up to reversal. Roles are scored as precision and recall of the predicted control and datapath cells. The hierarchy similarity compares the clusters of the module paths to the committed pieces with the Dice score $F(g,h) = 2|g \cap h|/(|g| + |h|)$, $\text{HS}_{gt} = \sum_g |g| \max_h F(g,h) / \sum_g |g|$ and its converse $\text{HS}_{l1}$, and reports their harmonic mean.
7) Aggregate. loki score writes one score.json per run and loki score-all the floor-by-kind tables, the reliability curve of $p_{\text{comp}}$ with its ECE, and, with –compare, the ablation against –features none.
8) Naming neutrality. The renamed twin of the dataset is run and scored against the original-named key; the base and renamed score.json must be byte-identical.

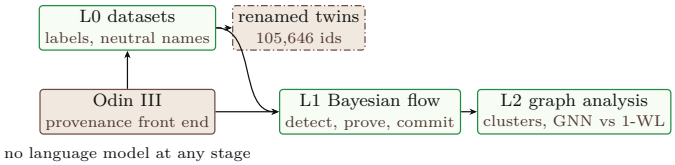



Fig. 1. The Loki III pipeline. L0 compiles the benchmarks through Odin III into provenance-tracked, labelled, anonymised datasets (and their renamed twins). L1 proposes and proves structures in a Bayesian flow that commits into the same IR. L2 reads the graphs back and scores clustering and a GNN against a 1-WL baseline. No stage has a language model.

Of the Gate-Breaker metrics, the AST and CFG similarities are used on f3 as in the paper: AST similarity is the mean, over the subtrees of the reference, of the best Jaccard similarity to a subtree of the candidate (depth 3, node kinds normalised); Gate-Breaker computes both on functionally correct samples only, Loki's are unconditional; CFG similarity is the node kernel over shortest-path histograms of the control-flow graphs. They are computed twice per design, for the gate netlist (L0-stop) and for design.v (L1-stop), and the L1 delta is their difference. The success rate and Success@$k$ need samples; a deterministic flow has one attempt per design, so they are reported only for the language-model replication (Section V-E). Formal final_equiv does not play the role of the testbench: it compares the flattened final design with the input netlist, so it shows that the rewrite preserved the netlist (it is trivial on f3, where L1 commits nothing), not that the design is correct against the reference RTL. Loki has no behavioural pass rate against the reference.

L2, step by step. (1) The same split applies. (2) loki analyze cluster writes a partition per method and design, with its modularity and NMI/ARI against module_path and construct_type; the table reports the mean over designs. (3) loki analyze gnn fits the vocabulary and the model on the training designs and writes metrics.json: accuracy and macro-F1 on the training and test nodes for the GNN, the majority class, the feature-only logistic regression and the 1-WL baseline, and beats_wl. (4) The neutrality check re-runs both on a permuted copy of every identifier.

### F. One design through the pipeline

The steps below follow one design, the 4-bit comparator micro bm_DL_4_bit_comparator (three comparison outputs), from source to score. Each step names the file or code that defines it.

1) HDL to Odin III with provenance. read_verilog and the common front elaborate the design. Every

TABLE II
The Gate-Breaker evaluation and the Loki evaluation, step by step.

| Step | Gate-Breaker | Loki |
|---|---|---|
| Input | cmos6 gate netlist of a VerilogEval reference, internal nets renamed n_⟨SHA-1 prefix⟩, ports kept | a floor netlist (f1, f2, f3) with names n$k$; for the neutrality check a renamed twin |
| Producer | a language model, $K$ samples per design | the L1 flow, one deterministic run per design (fixed seed); no sampling |
| Correctness | simulate the extracted RTL against the testbench: success rate, Success@$k$ | equiv-check of the flattened final design against the input netlist: final_equiv (formal; on f1 hard cells are cut points and only soft logic is checked). A self-consistency check of the rewrite, not correctness against the reference RTL; Loki has no behavioural pass rate against the reference |
| Match to the truth | AST and CFG similarity to the reference RTL | each counted hypothesis against the L0 answer key: Jaccard $\geq 0.5$, one-to-one; precision, recall, F1, average precision per kind and floor; hierarchy similarity; AST and CFG of design.v against the reference on f3, unconditionally (Gate-Breaker's are computed on functionally correct samples only) |
| Aggregate | per complexity quartile | micro-averaged per kind and floor over the test split; AST and CFG per quartile of f3 node count |

IR object keeps its source spans (Node.sources(), the only provenance the binding exposes). Before mapping, Loki snapshots every word-level cell (type, parameters, spans) and every instance site.

2) Map to the floor. lower keeps each word cell's exact span on every gate it creates; abc –lut 6 gives each LUT the union of the spans of its cone. A LUT that merges the three comparators therefore lists all three spans.
3) Join and label. A final cell is matched to a word cell when the word's span set is a subset of the cell's. The primary match gives construct_type (here $eq, $gt, …); a cell with several matches is flagged (n_construct_cells above one) and a cell with none stays unknown: a label is an answer key, never a guess. The module path comes from the instance spans, bus name and bit index from net names and aliases (AeqB[0]), and the datapath or control role from a rule table.
4) Anonymise and write the L0 files. Nets, cells and port bits get names n$k$ from one counter. Loki writes netlist.blif and netlist.v, graph.json (nodes with features and labels, edges), labels.csv (one row per node), words.json (the word-level connectivity the scorer needs) and key.json. Only key.json holds original names; no analysis reads it.
5) Verify. Odin III's equiv-check compares the written BLIF with the floor's own netlist and the anonymised netlist with the original; both must read equivalent before the floor counts as built.
6) L1: normalise. loki re reads netlist.blif (never key.json) into a fresh Odin III design, level 0, in which every logic cell is a function (a LUT by its truth table, a gate by its library function). On f1 with hard cells the unmap pass runs on the working copy and its facts feed the hard detector.
7) L1: hypothesis rounds. Each round runs the detectors, scores every hypothesis (Section III-C), proves the ones worth a proof, resolves conflicts and commits the confident ones (Algorithm 1); the next round sees the committed level.
8) L1: commit and output. A committed hypothesis is rebuilt as a word cell (raise) wrapped in a module (submod) tagged with its kind, confidence and recovered function. At the fixed point Loki writes design.v, overlay.json (every hypothesis with its evidence) and report.json, and checks the final design against the input with equiv-check.
9) L2. loki analyze reads graph.json only: structural features, four clusterings and a GNN (Section III-D).
10) Score. loki score compares the overlay with the answer key (Section III-E); loki score-all aggregates every run into per-kind tables.

In this micro the f3 comparator, as re-synthesised by the minimal .v front, has 26 logic cells (L1 spec, Section 3); the f2 LUTs carry the comparator spans that the join reads back.

## IV. The VerilogEval benchmark and its structure

### A. VerilogEval spec-to-rtl as the Gate-Breaker benchmark

Gate-Breaker evaluates on VerilogEval [2], taking the spec-to-rtl variant: 156 problems, each a natural-language prompt, a reference solution (*_ref.sv, module RefModule) and a testbench (*_test.sv). The Gate-Breaker flow synthesizes the reference to a cmos6 gate netlist, hands the netlist to the model, and checks the model's RTL against the testbench. Loki uses the same 156 references as L0 sources and reads the reference module as the design; the prompts and testbenches are not inputs to L0, L1 or L2. The references are SystemVerilog, read through Odin III's slang adapter (Section V-A).

B. The complexity score and its quartiles

Gate-Breaker groups the problems by a complexity score (its Eq. 2), a weighted sum of counts of HDL constructs:

$$C = 1000\,c_{\mathrm{Seq}} + 200\,(c_{\mathrm{FG}} + c_{\mathrm{F}}) + 20\,c_{\mathrm{Mo}} + 15\,(c_{@A} + c_{\mathrm{B}}) + 10\,(c_{\mathrm{M}} + c_{\mathrm{Ca}}) + 5\,c_{\mathrm{E}} + 3\,c_{\mathrm{A}} + 2\,c_{\mathrm{C}}. \quad (3)$$

The paper names the terms but does not define them, so the implementation we reuse (complexity.py of the second Gate-Breaker replication) fixes the reading: $c_{\mathrm{Seq}}$ counts clocked always blocks (a posedge or negedge in the sensitivity list), $c_{\mathrm{FG}}$ for-generate constructs, $c_{\mathrm{F}}$ for loops, $c_{\mathrm{Mo}}$ module instantiations, $c_{@A}$ all always blocks, $c_{\mathrm{B}}$ declarations wider than one bit, $c_{\mathrm{M}}$ multiplexers (ternary operators plus non-default case items), $c_{\mathrm{Ca}}$ case statements, $c_{\mathrm{E}}$ arithmetic operators (+ - * / %), $c_{\mathrm{A}}$ continuous assignments and $c_{\mathrm{C}}$ concatenations. A clocked block weighs 1000 on its own, so a design in which the counter recognises one scores at least 1000, which is Q3 or Q4.

The quartiles are the paper's absolute bins (its Table II), not rank quartiles of the benchmark: Q1 is 3 to 23, Q2 24 to 111, Q3 112 to 1064 and Q4 1065 to 1326. The 156 references fall 38, 39, 40 and 39 into Q1 to Q4, close to the paper's 39 per quartile. The 150 designs that build on every floor are split 38, 39, 37 and 36: the six missing ones are Prob031, Prob046 and Prob145 (Q3) and Prob078, Prob151 and Prob156 (Q4). Table III gives the counts.

Deviation: pyverilog AST against a regex fallback. The score is computed on the pyverilog AST of the first module when the file parses, and by regular-expression counts of the same terms otherwise, with the method recorded per design. Because the references are SystemVerilog, pyverilog parses only 60 of the 156; the other 96 use the regex counts. The split is skewed by quartile (Table III): 36 of Q1 and 23 of Q2 are AST-scored, but 1 of Q3 and none of Q4. The regex counts operator characters and keywords textually, so they differ from an AST count (the observed Q4 range, 1069 to 3259, exceeds the paper's bin top of 1326 for this reason or because the problems differ; we do not separate the two). The regex also never counts always_comb, always_ff or always_latch blocks, which 34 references use, so c@A and cSeq are undercounted in the 96 regex-scored designs, most of Q3 and Q4; the quartile split and the feature trends below rest on approximate scores. The 156 scores equal those of the replication that computed them first, exactly, so the quartiles here and in Table XII are the same partition.

C. The feature matrix

Gate-Breaker characterises its benchmark by the HDL structures it contains. We build the same kind of matrix for the 150 built designs, from the f3 floor of each, with rules fixed in advance (summarize.py): arithmetic counts the word cells of type $add $sub $mul $div $mod $neg $pos $pow $alu $macc; select/mux those of type $mux $pmux $bmux $demux; storage the register words ($dff, $dffe, $adff, $adffe, $sdff, $sdffe, $sdffce, $dffsr, $aldff, $dlatch, $adlatch); FSM is 1 when Odin III's fsm pass marked any word; control is 1 when any label row has role control or mixed; and wide buses counts the distinct bus names with a bit index of at least 1. The matrix uses one symbol per cell: ✗ for count 0, ✓ for one, ✓✓ for two or more; FSM and control are 0/1 flags, so they show ✗ or ✓ only.

The bus column is buses_wide, not buses. The latter counts every named signal, including single-bit ports, so it is at least 1 for all 150 designs and at least 2 for 147, and would put a check mark in every row. The wide count is the quantity a "multi-bit bus" row means; both columns are committed in features.csv.

Table IV shows 12 designs, three per quartile at fixed ranks by size, so the sample is not hand-picked; Table V counts, per quartile, the designs that show each feature. The storage trend is partly tautological: a clocked block adds 1000 to the score (Eq. 3) and storage counts register words, so storage rises with the quartile by construction; the other trends are conditioned on the score as well. Storage is absent from Q1 and Q2 (0 of 77 designs) and present in 31 of 37 Q3 and all 36 Q4 designs. The other features grow with the quartile: select/mux 8% in Q1 to 56% in Q4, control 3% to 83%, FSMs 0 of 77 below Q3 and 15 of 36 in Q4. Arithmetic is rare throughout (21 of 150, 14%), so the benchmark leans on control, selection and storage more than on arithmetic datapaths. Wide buses are common from Q2 up (90%, 59%, 100%) but nearly absent in Q1 (8%), whose members mostly have only single-bit signals. The FSM flag is Odin III's detection, not a reading of the source: Prob107_fsm1s in Table IV has storage and control cells but no FSM mark.

D. Benchmark size per technology mapping

Table VI gives the size of the 150 designs on each floor. The averages hide a heavy tail. Over all designs the mean node count is 163.5 on f1-k6, 111.4 on f2 and 362.0 on f3, but the medians are 14.0, 13.5 and 25.0: most references are small (125 of 150 have at most 100 nodes at f3) and a few are large (9 exceed 1000; the largest, Prob144_conwaylife, has 15,487). The gate mapping multiplies the node count of the LUT mapping by about 3 on the mean and 1.9 at the median (362.0 against 111.4 on average, 25.0 against 13.5 at the median), as expected when a LUT of up to six inputs becomes a network of cells from a library of one- and two-input gates (BUF, INV, NAND2, NOR2). f1-k6 sits above f2 on average and in Q3 and Q4 (109.4 against 77.4; 453.9 against 269.8) while the two coincide in Q1 and Q2 (6.3 and 6.1; 100.2 and 100.0): the f1-k6 floor keeps hard cells (21 designs hold hard adders) where f2 spreads arithmetic over LUTs, which is consistent with the excess but is not isolated here. By quartile on f3, the mean grows from 9.9 nodes in Q1 to 972.1 in Q4, but the Q4 median is only 51.5: the complexity score tracks the number and kind of

TABLE III
The VerilogEval spec-to-rtl references by Gate-Breaker complexity quartile (absolute bins of Table II; score range observed in the 156 refs; pyverilog AST scores against regex-fallback scores; built = designs with all three floors).

| Quartile | Bin | Observed | Refs | AST/regex | Built |
|---|---|---|---|---|---|
| Q1 | 3–23 | 3–23 | 38 | 36/2 | 38 |
| Q2 | 24–111 | 24–90 | 39 | 23/16 | 39 |
| Q3 | 112–1064 | 120–1060 | 40 | 1/39 | 37 |
| Q4 | 1065–1326 | 1069–3259 | 39 | 0/39 | 36 |
| All | | | 156 | 60/96 | 150 |

TABLE IV
Benchmark feature matrix, a sample of 12 of the 150 designs (the designs at ranks $\lfloor n/6 \rfloor$, $\lfloor n/2 \rfloor$ and $\lfloor 5n/6 \rfloor$ by f3 node count inside each quartile). ✗: feature absent; ✓: one; ✓✓: more than one; FSM and control are 0/1 flags. Features are read from the f3 floor (Section IV-C).

| Design | Q | f3 nodes | Arith | Sel | Stor | FSM | Ctl | Bus |
|---|---|---|---|---|---|---|---|---|
| Prob028_m2014_q4a | Q1 | 3 | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Prob036_ringer | Q1 | 8 | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Prob091_2012_q2b | Q1 | 17 | ✗ | ✗ | ✗ | ✗ | ✗ | ✓ |
| Prob125_kmap3 | Q2 | 15 | ✗ | ✗ | ✗ | ✗ | ✗ | ✗ |
| Prob015_vector1 | Q2 | 32 | ✗ | ✗ | ✗ | ✗ | ✗ | ✓✓ |
| Prob123_bugs_addsubz | Q2 | 166 | ✓✓ | ✓ | ✗ | ✗ | ✗ | ✓✓ |
| Prob107_fsm1s | Q3 | 11 | ✗ | ✓ | ✓ | ✗ | ✓ | ✗ |
| Prob117_circuit9 | Q3 | 28 | ✓ | ✓ | ✓✓ | ✗ | ✓ | ✓ |
| Prob045_edgedetect2 | Q3 | 89 | ✗ | ✗ | ✓✓ | ✗ | ✗ | ✓✓ |
| Prob128_fsm_ps2 | Q4 | 30 | ✗ | ✗ | ✓ | ✓ | ✓ | ✓✓ |
| Prob140_fsm_hdlc | Q4 | 52 | ✗ | ✗ | ✓ | ✓ | ✓ | ✓ |
| Prob115_shift18 | Q4 | 1121 | ✗ | ✓✓ | ✓✓ | ✗ | ✓ | ✓✓ |

HDL constructs, not the netlist size. For scale, the micro suite's f3 floors hold 50,047 nodes over 65 designs, a mean of about 770.

## V. Results

### A. L0 on VerilogEval: the provenance ceiling

The provenance experiment is run on the 156 VerilogEval spec-to-rtl reference designs (Section IV) next to the micro suite. The question is the one L0 exists to answer: how much of the reference does the answer key reconstruct, per floor, with no recovery step at all. The experiment builds the three floors f1-k6, f2 and f3 for every reference (f1-earch is not built; one architecture is enough because f2 and f3 do not depend on it).

Coverage of the build. Of the 156 references, 150 build on all three floors, and loki check reports zero errors on the assembled dataset. Two are skipped by Odin III's SystemVerilog reader: Prob151_review2015_fsm and Prob156_review2015_fancytimer (both Q4), with ODIN3_ERR_PARSE ... an explicit 4-bit cast is not supported. Four fail Loki's own equivalence gate and are recorded as failed rather than forced in: Prob031_dff (the reference initialises q to 1'hx; the writer-versus-Odin III BLIF check reports differs on f1 and f2, while f3 was equivalent) and Prob046_dff8p, Prob078_dualedge and Prob145_circuit8, whose negedge flip-flops make equiv-check return an error status for both comparisons. One design, Prob144_conwaylife, was killed by Wyrd's default 2 GB step memory limit (peak 4.3 GB) and built after a rerun with an 8 GB limit. Reading .sv sources needed two fixes in Loki (SystemVerilog is read with –nolatches only, since –vtr-compat is refused for it, and the slang adapter is registered with the process); both are in loki/odin.py and apply to the experiment run.

The ceiling. On f3 the 150 designs have 54,306 gate nodes. Every node has a label row, a cell_type and an HDL span: the label, cell_type and HDL coverage have geometric mean 1.000 over designs. Label coverage is rows over nodes, so it shows that a row exists, not that its fields are informative; unmatched cells stay unknown. The 150 module scopes of the 150 reference modules are all present (module coverage 1.000). Signal coverage, the recovered leaf signal names over the signals the reference declares, has geometric mean 0.986 (arithmetic mean 1.029): 120 designs are at exactly 1, 28 are below it, and 2 are above it, where flattened array elements add names. The micro suite shows the same shape with a slightly lower ceiling (Table VII): 99.1% of module scopes and 91.7% of signal names (an aggregate ratio, where VerilogEval's 841 over 745 is inflated by those two designs; capped at each design's declared count it is 715 over 745, 96.0%). The two sets agree on the message of Section V-C: every node has a label row, a cell_type and an HDL span, and what the mapping erased is the signal names, not the label rows. (Label coverage counts rows, not informative values: a cell that no rule matched keeps the value unknown.)

Caveats, reported with the numbers. (i) –nolatches turns inferred-latch paths into don't-care and lower –async sync turns asynchronous resets into synchronous clears, so a built netlist can differ from the reference's behaviour even though Loki's own checks (netlist against its anonymised twin and against Odin III's writer) pass; the build gate guarantees internal consistency, not equality with the reference. (ii) The reference module, instance and signal counts come from pyverilog where it parses the reference (130 of the 150) and from a regex fallback otherwise (20); the regex counts are approximate and the reference signal count is the weakest number in this section. (iii) The same pyverilog limit decides the complexity quartiles (Section IV-B). (iv) A train/test split by design is fixed by the manifest rule (117 train, 33 test on f3); it is not used here, where no model is fitted.

### B. L1 recovery: what the flow already reverse engineered, and what it erased

Table VIII reports L1 on the renamed floors (identical to base, as the neutrality check requires). The head-line result is the hard detector: hard adders recover 67 of 67 with precision 0.985, hard multipliers 30 of 30, because the synthesis flow itself already reverse engineered them and unmap says so. The rest is the honest difficulty: on f1 the calibrated macro-F1 is 0.354 (counted 271, matched

TABLE V
Designs showing each feature, per complexity quartile: count (percent of the quartile). The last column counts designs with two or more instances of the feature (always 0 for the 0/1 flags). Wide buses are distinct multi-bit buses; a design with only single-bit ports has none.

| Feature | Q1 ($n$=38) | Q2 ($n$=39) | Q3 ($n$=37) | Q4 ($n$=36) | All ($n$=150) | $\geq 2$ |
|---|---|---|---|---|---|---|
| Arithmetic | 2 (5%) | 4 (10%) | 7 (19%) | 8 (22%) | 21 (14%) | 9 |
| Select/mux | 3 (8%) | 10 (26%) | 12 (32%) | 20 (56%) | 45 (30%) | 12 |
| Storage | 0 (0%) | 0 (0%) | 31 (84%) | 36 (100%) | 67 (45%) | 12 |
| FSM | 0 (0%) | 0 (0%) | 1 (3%) | 15 (42%) | 16 (11%) | 0 |
| Control | 1 (3%) | 4 (10%) | 15 (41%) | 30 (83%) | 50 (33%) | 0 |
| Wide buses | 3 (8%) | 35 (90%) | 22 (59%) | 36 (100%) | 96 (64%) | 55 |

TABLE VI
Benchmark size per technology-mapping floor, 150 designs: mean / median node count (top), and per complexity quartile (bottom; nodes). Nodes are all graph nodes of the floor netlist: primary inputs and outputs, constants, cells.

| All 150 designs | f1-k6 | f2 | f3 |
|---|---|---|---|
| Nodes | 163.5 / 14.0 | 111.4 / 13.5 | 362.0 / 25.0 |
| Cells | 114.6 / 5.0 | 62.5 / 4.0 | 313.2 / 13.5 |
| Nets | 218.3 / 12.0 | 93.5 / 10.0 | 344.2 / 21.0 |
| **Nodes by quartile** | **f1-k6** | **f2** | **f3** |
| Q1 ($n$=38) | 6.3 / 5.0 | 6.1 / 5.0 | 9.9 / 8.0 |
| Q2 ($n$=39) | 100.2 / 26.0 | 100.0 / 26.0 | 238.2 / 32.0 |
| Q3 ($n$=37) | 109.4 / 18.0 | 77.4 / 13.0 | 260.6 / 28.0 |
| Q4 ($n$=36) | 453.9 / 26.5 | 269.8 / 26.5 | 972.1 / 51.5 |

TABLE VII
The L0 ceiling on f3, micro suite and VerilogEval references. Both columns are counts over all built designs; the last row is the per-design coverage of declared signals.

| | Micro | VerilogEval |
|---|---|---|
| Source designs | 68 | 156 |
| Built on f3 | 65 | 150 |
| Refused by the reader | 3 | 2 |
| Failed equivalence gate | 0 | 4 |
| Gate nodes, each with a label row | 50,047 | 54,306 |
| Rows with cell_type and HDL span | 100% | 100% |
| Module scopes / reference modules | 112 / 113 | 150 / 150 |
| Leaf signal names / declared signals | 611 / 666 | 841 / 745 |
| Per-design signal coverage: median | 100% | 100% |
| minimum | 55.6% | 75.0% |
| geometric mean | – | 0.986 |

TABLE VIII
L1 recovery on the renamed floors, per kind. Precision/recall/F1 against the L0 ground truth; calibrated (micro-fit) columns. The hard kinds recover from unmap facts; the rest must be re-found from the gates.

| Kind (f1) | Precision | Recall | F1 |
|---|---|---|---|
| hard_add | 0.985 | 1.000 | 0.993 |
| hard_mul | 1.000 | 1.000 | 1.000 |
| ff_en | – | – | 0.364 |
| register | – | – | 0.021 |
| mux | – | – | 0.018 |
| mux_bit | – | – | 0.081 |
| macro-F1 (all kinds) | | | 0.354 |
| **Kind (f2)** | | | |
| adder | 1.000 | 0.134 | 0.237 |
| subtractor | 1.000 | 0.074 | – |
| mux_bit | – | – | 0.100 |
| adder_cin | – | – | 0.029 |
| macro-F1 (all kinds) | | | 0.108 |

107, ground truth 688), with ff_en at 0.364, registers at 0.021 and muxes at 0.018; on f2 it is 0.108 (91/19/686), with adders at F1 0.237 (precision 1.000, recall 0.134) and most other kinds near zero. Without calibration the flow proposes far more and confirms little: f1 macro-F1 0.202 with 1,128 counted (323 of them declined-adder proposals), f2 0.075. Calibration is honest about itself: ECE 0.217 calibrated against 0.650 uncalibrated.

### C. f3: recovery from the cmos6 gate netlist

Gate-Breaker's own input is a cmos6 gate netlist, so we built the f3 floor (65 of 68 micro designs; the gate reader refuses three hard-block sources) and asked the same questions of it. The L0 answer key is complete at this level in that all 50,047 gate nodes carry a label row, a cmos6 cell_type and an HDL span. What provenance reconstructs of the reference is nearly complete but not total: 112 module scopes against 113 module definitions (99.1%; 54 of 65 designs complete) and 611 distinct leaf signal names against 666 declared signals (91.7%; median design 100%, minimum 55.6%). The module gap is reference-side (generate branches and unused submodules never elaborate); the signal gap is synthesis-side (nets that mapping created or merged have no original name left). The ceiling is consequently what the mapping preserved, not the source RTL.

Table IX reports L1 on the 15 f3 test designs (23,098 gate nodes), base and renamed, with a calibration fitted on the 50 train designs' f3 floors. The result is the strict one: no construct is counted (0 of 144 ground-truth instances; macro-F1 0.0) and no hypothesis is committed at all. The f3 priors are strongly negative because the gate-level detectors are imprecise on the train floors (comparator $-9.24$ from 31,036 proposals with 2 true), so the best calibrated confidence in any test design is a bus at $p = 0.764$, below the 0.9 commit threshold, and the L1-stop Verilog is byte-identical to the input netlist in 15 of 15 designs. Without calibration the detectors count 1,166 constructs for 2 matches (macro-F1 0.038; one adder, one register) and 2,835 buses for 0 matches. Buses are proposed

TABLE IX

f3 L1 per kind on the 15 test designs. The calibrated columns are the protocol's setting; the –calib none columns are a diagnostic of what the detectors propose before calibration prunes it. A kind with no counted prediction has no F1 and is left out of the macro average.

| | | calibrated | | | –calib none | | | |
|---|---|---|---|---|---|---|---|---|
| Kind | GT | counted | matched | F1 | counted | matched | precision | F1 |
| adder | 17 | 0 | 0 | – | 1 | 1 | 1.000 | 0.111 |
| adder_cin | 17 | 0 | 0 | – | 0 | 0 | – | – |
| comparator | 20 | 0 | 0 | – | 14 | 0 | 0.000 | 0.000 |
| eq_const | 16 | 0 | 0 | – | 1,141 | 0 | 0.000 | 0.000 |
| ff_srst | 2 | 0 | 0 | – | 7 | 0 | 0.000 | 0.000 |
| hard_add | 17 | 0 | 0 | – | 0 | 0 | – | – |
| hard_mul | 11 | 0 | 0 | – | 0 | 0 | – | – |
| mux | 7 | 0 | 0 | – | 0 | 0 | – | – |
| mux_bit | 7 | 0 | 0 | – | 0 | 0 | – | – |
| register | 22 | 0 | 0 | – | 3 | 1 | 0.333 | 0.080 |
| subtractor | 8 | 0 | 0 | – | 0 | 0 | – | – |
| macro-F1 | 144 | 0 | 0 | 0.000 | 1,166 | 2 | | 0.038 |
| buses | 120 | 2,569 | 0 | 0.000 | 2,835 | 0 | 0.000 | 0.000 |
| datapath role | 20,541 | 20,541 | 20,541 | 1.000 | 20,541 | 20,541 | 1.000 | 1.000 |
| control role | 0 | 0 | 0 | – | 0 | 0 | – | – |

but never reconstructed: 2,569 counted bus hypotheses against 120 ground-truth buses, 0 matched, so no width or bit order is recovered. The role detector labels every logic cell datapath, which is 100% correct only because the f3 answer key holds no control cells (0 of 50,047; f1 has 4): control-versus-data is not answerable from f3 provenance, and the datapath F1 of 1.000 over 20,541 cells is vacuous. The hard detector stays f1-only by design (unmap runs only on f1 floors with hard cells), so the 28 hard ground-truth instances are unreachable here. Base and renamed score.json digests are identical (9368cfc26303): naming contributes no decision bit at f3 either.

The AST and CFG deltas (Table X) follow from the zero-commit result: with the f3 calibration L1-stop is byte-identical to L0-stop, so both metrics have delta exactly 0.0000 in every quartile (AST over 15 designs; CFG over the 13 whose CFG the probe can finish, 12,000 nodes being its practical $O(n^2)$ limit). The uncalibrated diagnostic, where L1 commits 1,535 hypotheses in 11 of 15 designs, moves the means by +0.0111 (AST) and −0.0075 (CFG), with mixed per-quartile signs; the commits are mostly wrong eq_const and bus pieces, so this is not evidence about recovery quality. A flatten control (Odin III proc; flatten, which also lowers the committed processes) inlines the committed hierarchy and raises both metrics over the gate netlist: AST 0.4576 against 0.4044 (+0.0533, all 15 designs) and CFG 0.6941 against 0.6382 (+0.0559, the 12 designs the probe can score). Gate-Breaker's Algorithms 2 and 3 read a single module, while Loki's L1 commits a hierarchy, so a recovery that produces hierarchy is systematically penalised until it is flattened: for a netlist-to-RTL benchmark this is a metric limitation, not a property of the recovery.

## D. L2: clusters that carry the hierarchy, and a GNN that beats 1-WL

Table XI scores the four clusterings against the known module hierarchy. Louvain leads (modularity 0.4220, NMI 0.1360); k-means and affinity propagation do not find the

TABLE X

f3 AST (Gate-Breaker Algorithm 2, depth 3, pyverilog) and CFG (Algorithm 3, odin3 probe) similarity against the micro reference RTL, L0-stop (gate netlist) against L1-stop (design.v); quartiles by f3 node count. With the f3 calibration L1-stop equals L0-stop, so every delta is 0.0000; the table therefore shows the L0 values and the diagnostic (–calib none) deltas.

| Quartile | AST L0 | AST L1 | ΔAST | CFG L0 | CFG L1 | ΔCFG |
|---|---|---|---|---|---|---|
| 0 (4 designs) | 0.4428 | 0.4428 | 0.0000 | 0.7909 | 0.7909 | 0.0000 |
| 1 (4) | 0.3765 | 0.3765 | 0.0000 | 0.4224 | 0.4224 | 0.0000 |
| 2 (4) | 0.3969 | 0.3969 | 0.0000 | 0.7013 | 0.7013 | 0.0000 |
| 3 (3 AST / 1 CFG) | 0.4003 | 0.4003 | 0.0000 | 0.7507 | 0.7507 | 0.0000 |
| mean (15 / 13) | 0.4044 | 0.4044 | 0.0000 | 0.6468 | 0.6468 | 0.0000 |
| diagnostic –calib none (L1 commits hierarchy) | | | | | | |
| 0 (4) | 0.4428 | 0.4277 | −0.0151 | 0.7909 | 0.7776 | −0.0133 |
| 1 (4) | 0.3765 | 0.3756 | −0.0009 | 0.4224 | 0.4099 | −0.0124 |
| 2 (4) | 0.3969 | 0.4542 | +0.0573 | 0.7013 | 0.7014 | +0.0001 |
| 3 (3 AST / 1 CFG) | 0.4003 | 0.4007 | +0.0004 | 0.7507 | 0.7561 | +0.0055 |
| mean | 0.4044 | 0.4155 | +0.0111 | 0.6468 | 0.6394 | −0.0075 |

TABLE XI

L2 clustering against the known module hierarchy (68 designs, mean over the renamed floors, identical to base). Modularity and NMI(module_path) per method.

| Method | Modularity | NMI |
|---|---|---|
| Louvain | 0.4220 | 0.1360 |
| Spectral | 0.3620 | 0.1378 |
| Affinity propagation | -0.2264 | 0.2708 |
| k-means | -0.2912 | 0.3220 |

hierarchy (negative modularity), which is itself a result: graph structure alone, without the labels, favours the community methods. 271 of 272 partition records are identical between base and renamed floors; the one difference exposed a determinism defect in the spectral path (repeated runs on the same base design give modularities -0.0079, -0.0410 and 0.0252), reported as an open issue rather than hidden.

The GNN result answers the paper's question directly. A node-only GraphSAGE loses to the 1-WL baseline on real micros (macro-F1 0.2413 against 0.2711); combining node, graphlet, WL and NPN features turns it around (0.3545 against 0.2711 on the earlier real-micro run), and the full labelled run of this experiment gives test macro-F1 0.4241 against 0.2717 (accuracy 0.8863 against 0.8492, 3,932 test nodes across 13 designs). The features carry the recovery, not the message passing alone.

## E. The Gate-Breaker metrics

Gate-Breaker's metrics compare a recovered RTL against a reference: Success Rate, Success@$k$, AST similarity and CFG similarity. Loki recovers structure, not a new RTL text, so the direct comparison is the replication: the Gate-Breaker harness itself, run on the simple benchmarks with the only language model available on the host, gave success rate 0.933 on Q1 and 0.829 on Q2, Success@10 1.000 in both quartiles, AST similarity 0.576 and 0.444, CFG similarity 1.000 and 0.736, with 169 of 174 extracted candidates functionally correct – a first replication on the fallback micro set, not the paper's VerilogEval set.

TABLE XII
Gate-Breaker metrics on the VerilogEval quartiles (one run, K=10, 4 workers, 180 s per call, 7,440 s wall clock, 740 calls; deltas vs the paper's O3-Mini in the text).

| Quartile ($n$) | SR | S@1 | S@5 | S@10 | AST | CFG |
|---|---|---|---|---|---|---|
| Q1 (36) | 0.903 | 0.944 | 1.000 | 1.000 | 0.894 | 0.942 |
| Q2 (37) | 0.778 | 0.811 | 1.000 | 1.000 | 0.761 | 0.890 |

The VerilogEval replication, on the quartiles the paper used, is reported in Table XII with the AST and CFG kernels recomputed with common-width feature vectors (the first run padded per graph and truncated in the zip, which the v1.1.0 CFG probe found; those columns are not comparable to it).

Seventy-four of the 78 Q1+Q2 designs are usable (three excluded because their synthesized netlists fail the official testbench self-check, one because its 241 KB prompt exceeds the OS argument limit); every scored design is solved by attempt 5. The deltas against the paper's O3-Mini: Q1 success rate -0.092, AST -0.030, CFG -0.011; Q2 success rate -0.039, Success@5 +0.125, Success@10 +0.111, AST -0.164, CFG +0.083. The Q1 gap is harness no-output, not wrong RTL; the Q2 AST gap is truth-table and case references against the model's boolean expressions. Two caveats: the host's only model path routes through a gateway that aliases a DeepSeek backend as Claude Opus (so the model column is not verifiably the paper's comparison class), and the CLI exposes none of the paper's sampling flags ($T$=0.85, $p$=0.95). The CFG column uses the common-width node kernel (the odin3 CFG probe); the AST column uses the paper's pyverilog Algorithm 2 of Gate-Breaker.

What the two lines of work measure is complementary. The language model recovers functional behaviour from the gates and names nothing reliably; Loki recovers the structures the flow preserved (hard blocks, the hierarchy, the signal roles its features see) and proves what it claims. The Gate-Breaker future work – bus naming, datapath/control classification, partitioning – is exactly L2's table, and its numbers are the baseline a model-assisted Loki would have to beat.

## F. Round 2: five follow-up questions

The first round left five questions open. Each carries a hypothesis and a decision rule; the thresholds were set by the implementer during the round and are not time-stamped before the data (Appendix A); a later round should fix them in a commit before running. All five are compute-only: no language-model call was made, and the language-model figures are the round-1 Gate-Breaker replication, read again. Every number in this subsection is a macro generated from the summary tables of experiments/round2.

RQ-L1: is the f3 failure a vocabulary problem? The L1 detectors read LUT covers, and the f3 netlist is a cmos6 gate netlist. H1: re-mapping the same gates into LUT6 recovers close to the f2 rate. We read each f3 netlist.blif into Odin III, run abc –lut 6, and give the result to loki re as an f2 netlist. Odin III drops the cell names in abc, so each of the 8,712 LUTs is expanded back to the gates it covers: a bit-parallel simulation signature matches its output to the f3 net of the same function, and the cover is the cone of that net down to the matched inputs. 0.230 of the LUTs ABC resynthesised implement a function that is no f3 net; they take the support-contained share of their consumers' covers, and 153 are left with an empty cover. The expanded hypotheses are then scored against the unchanged f3 key, and the native f2 floor of the same 65 designs, with the same calibration, is the comparison (Table XIII). On the gate netlist the calibrated flow matches 0 of 116 instances in the 15 test designs. After the LUT6 re-map it matches 2 (native f2: 4), and over all 65 designs 14 against 16 of 580, a micro-F1 ratio of 0.87 in-sample (the 65 designs include the designs the f2 calibration was fitted on) against 0.51 on the held-out test split. The rule asked for a ratio of at least 0.8 on both splits, so H1 is inconclusive. The in-sample recovery is that of f2, on the same kinds (adders, subtractors, muxes), but the 15-design test split holds only a handful of matches (2 against 4), so its ratio is noise in either direction. The result is consistent with the f3 failure being the detectors' vocabulary and not missing information in the gates, but does not show it; the f2 level it would recover to is itself

TABLE XIII
Round-2 L1 (RQ-L1): recovery from the f3 gate netlist re-mapped to LUT6 against the native f2 floor, same 65 micro designs. test is the 15 held-out designs, all the 65. Calibration: the micro-fit f2 table (gate rows: the f3 table). Hard kinds excluded from counts and averages; macro-F1 averages the kinds with a counted prediction.

| Netlist | Calib. | Split | Runs | Counted | Matched | GT | micro-F1 | macro-F1 |
|---|---|---|---|---|---|---|---|---|
| f3 gate netlist | calibrated | test | 15 | 0 | 0 | 116 | 0.000 | 0.000 |
| | uncalibrated | test | 15 | 1,166 | 2 | 116 | 0.003 | 0.038 |
| f3 re-mapped to LUT6 | calibrated | test | 15 | 68 | 2 | 116 | 0.022 | 0.076 |
| | calibrated | all | 65 | 92 | 14 | 580 | 0.042 | 0.062 |
| | uncalibrated | test | 15 | 252 | 4 | 116 | 0.022 | 0.044 |
| | uncalibrated | all | 65 | 921 | 31 | 580 | 0.041 | 0.049 |
| f2 floor (native LUT6) | calibrated | test | 15 | 72 | 4 | 116 | 0.043 | 0.153 |
| | calibrated | all | 65 | 88 | 16 | 579 | 0.048 | 0.095 |
| | uncalibrated | test | 15 | 255 | 13 | 116 | 0.070 | 0.121 |
| | uncalibrated | all | 65 | 930 | 32 | 579 | 0.042 | 0.070 |

TABLE XIV
Operator-recovery F1 (RQ-L7) of the existing and new L1 runs: the multiset of recovered operators (class and operand width) against the operators of the L0 key, pooled over designs. type ignores the width, tw requires it.

| Run set | Designs | Key ops | Got ops | P | R | type-F1 | tw-F1 |
|---|---|---|---|---|---|---|---|
| f1-k6, calibrated | 68 | 468 | 169 | 0.657 | 0.237 | 0.349 | 0.204 |
| f1-k6, none | 68 | 468 | 964 | 0.233 | 0.481 | 0.314 | 0.166 |
| f2, calibrated | 68 | 468 | 21 | 0.810 | 0.036 | 0.070 | 0.061 |
| f2, none | 68 | 468 | 756 | 0.180 | 0.291 | 0.222 | 0.080 |
| f3 gates, calibrated | 15 | 108 | 0 | – | 0.000 | 0.000 | 0.000 |
| f3 gates, none | 15 | 108 | 1,161 | 0.009 | 0.102 | 0.017 | 0.008 |
| f3 as LUT6, calibrated | 65 | 463 | 19 | 0.895 | 0.037 | 0.071 | 0.062 |
| f3 as LUT6, none | 65 | 463 | 753 | 0.190 | 0.309 | 0.235 | 0.086 |
| VerilogEval f2, calibrated | 150 | 3,585 | 11 | 0.727 | 0.002 | 0.004 | 0.002 |
| VerilogEval f2, none | 150 | 3,585 | 1,151 | 0.196 | 0.063 | 0.095 | 0.042 |
| VerilogEval f3, calibrated | 150 | 3,585 | 2 | 1.000 | 0.001 | 0.001 | 0.001 |
| VerilogEval f3, none | 150 | 3,585 | 5,488 | 0.107 | 0.163 | 0.129 | 0.027 |

TABLE XV
AST and CFG similarity against operator F1 on the 15 f3 test designs (the f3-recovery runs). Calibrated and uncalibrated L1 tie on AST and CFG; the flatten control moves both and leaves the recovery record unchanged.

| Condition | AST | CFG | Ops recovered | type-F1 |
|---|---|---|---|---|
| L0-stop gate netlist | 0.4044 | 0.6468 | 0 | 0.0000 |
| L1 calibrated | 0.4044 | 0.6468 | 0 | 0.0000 |
| L1 uncalibrated | 0.4155 | 0.6394 | 1,161 | 0.0173 |
| L1 uncalibrated, flattened | 0.4576 | 0.6941 | 1,161 | 0.0173 |

TABLE XVI
Head-to-head on the VerilogEval designs the language model was scored on (Q1 and Q2, functionally correct samples; AST depth 3 and CFG as in Table XII). Loki columns: the output Verilog of the flow at that floor against the reference .sv; L0 is the netlist before L1.

| Metric | LLM (round 1) | f3 netlist (L0) | f3 L1 cal. | f3 L1 none | f2 L1 cal. | f2 L1 none |
|---|---|---|---|---|---|---|
| Q1 AST | 0.894 | 0.763 | 0.763 | 0.680 | 0.804 | 0.720 |
| Q1 CFG | 0.942 | 0.751 | 0.751 | 0.565 | 0.821 | 0.664 |
| Q2 AST | 0.761 | 0.460 | 0.460 | 0.438 | 0.459 | 0.448 |
| Q2 CFG | 0.890 | 0.743 | 0.743 | 0.679 | 0.787 | 0.764 |
| Q1+Q2 AST | 0.827 | 0.611 | 0.611 | 0.559 | 0.632 | 0.584 |
| Q1+Q2 CFG | 0.916 | 0.747 | 0.747 | 0.622 | 0.804 | 0.714 |

TABLE XVII
Loki on all 150 VerilogEval designs by complexity quartile (mean AST and CFG of the output Verilog against the reference; last column: operator type-F1 of the uncalibrated f2 run). Calibrated f3 equals the f3 netlist wherever L1 commits nothing.

| Quartile | $n$ | f3 AST | f3 CFG | f3 cal. AST | f3 cal. CFG | f2 cal. AST | f2 cal. CFG | op F1 |
|---|---|---|---|---|---|---|---|---|
| Q1 | 38 | 0.754 | 0.750 | 0.754 | 0.750 | 0.793 | 0.820 | 0.061 |
| Q2 | 38 | 0.455 | 0.741 | 0.455 | 0.741 | 0.455 | 0.784 | 0.075 |
| Q3 | 32 | 0.614 | 0.751 | 0.605 | 0.750 | 0.607 | 0.811 | – |
| Q4 | 21 | 0.505 | 0.768 | 0.505 | 0.768 | 0.507 | 0.870 | – |

TABLE XVIII
Evidence-family ablation (RQ-L4) on the 49 held-out f2 designs (16 micro, 33 VerilogEval). Each row refits the calibration on the 169 train designs with the family removed (its weight set to 0). The l2 family has no feature in loki re and is not applicable. The interval is a 95% design bootstrap of micro-F1 (hard kinds excluded); the next column is micro-F1 with the design ve_Prob153_gshare (about 1,300 of the 1,558 instances) left out.

| Calibration | Counted | Matched | micro-F1 | 95% interval | w/o gshare | macro-F1 | ECE | AUC |
|---|---|---|---|---|---|---|---|---|
| defaults (no calibration) | 1,708 | 26 | 0.0159 | 0.0036–0.0912 | 0.0553 | 0.113 | 0.841 | 0.437 |
| full calibration | 10 | 3 | 0.0038 | 0.0000–0.0444 | 0.0225 | 0.145 | 0.061 | 0.885 |
| without npn | 10 | 3 | 0.0038 | 0.0000–0.0444 | 0.0225 | 0.145 | 0.061 | 0.885 |
| without wl | 13 | 3 | 0.0038 | 0.0000–0.0435 | 0.0222 | 0.145 | 0.024 | 0.930 |
| without struct | 11 | 4 | 0.0051 | 0.0000–0.0578 | 0.0299 | 0.163 | 0.056 | 0.884 |
| priors only | 14 | 4 | 0.0051 | 0.0000–0.0574 | 0.0295 | 0.163 | 0.026 | 0.918 |

low.

RQ-L7: are AST and CFG blind to what Loki recovers? We score recovery with a word-level operator F1. The L0 key lists the word cells of the elaborated reference; a run's recovered operators are its counted hypotheses (committed, or $p_{\text{comp}} \geq 0.5$), mapped to an operator class and an operand width (add, sub, mul, eq, ne, lt, le, gt, ge, mux, register). Precision and recall are multiset overlaps, so the score does not depend on which cells carry an operator; type ignores the width and tw requires it. Table XIV rescores every run set. The floors separate cleanly, which neither AST nor CFG can show because both exist only on f3: type-F1 0.349 on f1 (the hard detector), 0.070 on f2 and 0.000 on the f3 gates. On the f3 test designs (Table XV) the calibrated and uncalibrated L1 runs tie on both metrics, AST by 0.0111 and CFG by $-0.0075$, while operator type-F1 moves by 0.0173; the flatten control raises AST by 0.0422 and CFG by 0.0547 and leaves the recovery record, hence the operator F1, unchanged (0.0000). By the rule H7 is supported by the letter of the rule (weak evidence). The tie the rule was written for, the calibrated flow against the L0-stop netlist (both add nothing to the text), does not separate: their operator type-F1 is 0 against 0. The verdict rests on the calibrated flow against the uncalibrated one, where the calibrated side recovers no operator at all, and on the flatten clause, which holds by construction. Two cautions: the separating level is small (the uncalibrated detectors count mostly wrong constructs), and the flatten comparison holds the record fixed by construction, because the operator F1 reads the recovery record and AST and CFG read the text. The operator F1 does not replace them; it is the measure that sees recovery, where they see only form.

RQ-L2: Loki against the language model on the same designs. Round 1 scored the language model on Q1 and Q2 of VerilogEval, and Loki's AST and CFG only on micro designs. We ran loki re on all 150 VerilogEval netlists at f2 and f3, calibrated (a calibration fitted on the 117 train designs of VerilogEval itself) and not, and scored the output Verilog (design.v) against the reference .sv with the round-1 AST and CFG code. 21 references that pyverilog cannot parse have no AST value for any candidate, and 6 f3 outputs exceed the CFG probe's size limit (l2_coverage.csv). On the designs both are scored on (Table XVI), Loki's f3 output has AST similarity 0.763 in Q1 and 0.460 in Q2 (common designs, $n = 36$ and 36) against the model's 0.894 and 0.761, and CFG similarity 0.751 and 0.743 against 0.942 and 0.890; every gap is significant by a paired Wilcoxon test (l2_paired.csv). The calibration was fitted on the train designs and these means cover all 150, so we also give the held-out test split alone (l2_testsplit.csv): on its 16 Q1 and Q2 designs both tools score on, AST is 0.705 for Loki against 0.893 for the model and CFG 0.802 against 0.927, the same direction. (The calibrated f3 flow commits in only 7 designs, so the fit can change few outputs.) Across the four quartiles (Table XVII) the f3 AST over all scored designs ($n = 38$, 38, 32, 21; a different, larger design set than the common rows above) reads 0.754, 0.455, 0.605 and 0.505, so it falls from Q1 to Q4 without being monotone. The 21 references that do not parse fall in Q1 to Q4 as 0, 1, 5 and 15, so Q3 and Q4 are scored on the designs that parse, and the apparent fall may partly reflect which designs those are. The CFG stays flat (0.750 to 0.768). The calibrated f3 flow commits almost nothing, so its output is the L0 netlist; the uncalibrated flow, which does commit, scores lower, not higher.

TABLE XIX
Graph sandwich (RQ-L5): Spearman correlation of the bracket width (candidate fraction of the cells) and of the committed fraction with the per-design F1; cell-level soundness against the union of the scored ground-truth footprints (must: committed cells; may: committed plus candidate cells).

| Run set | Designs | $\rho$ width | $p$ | $\rho$ must | must prec. | must rec. | may rec. |
|---|---|---|---|---|---|---|---|
| VerilogEval f2, uncalibrated | 90 | 0.087 | 0.417 | 0.485 | 0.840 | 0.240 | 0.625 |
| VerilogEval f2, calibrated | 90 | −0.203 | 0.055 | 0.819 | 0.821 | 0.025 | 0.188 |
| VerilogEval f2, calibrated, rp 0.1 | 90 | −0.346 | 0.001 | 0.819 | 0.821 | 0.025 | 0.447 |
| VerilogEval f3, uncalibrated | 90 | −0.124 | 0.243 | 0.168 | 0.792 | 0.347 | 0.727 |
| micro f2, uncalibrated | 48 | 0.162 | 0.272 | 0.219 | 0.879 | 0.268 | 0.364 |
| micro f2, calibrated | 48 | 0.052 | 0.725 | 0.844 | 0.932 | 0.021 | 0.132 |

TABLE XX
Structural prefilter (RQ-L5b), measured offline on the 210 cone_check calls of all round-2 runs (166 proved equivalent): the rules that prune calls without losing a proved construct, and the largest rule overall.

| Rule (kind/feature=bin) | Calls pruned | Fraction | Equivalent lost |
|---|---|---|---|
| adder/out_reads=no | 14 | 0.067 | 0 |
| subtractor/width=2 | 8 | 0.038 | 0 |
| subtractor/out_reads=no | 4 | 0.019 | 0 |
| adder/npn_sum=low | 3 | 0.014 | 0 |
| subtractor/npn_carry=all | 3 | 0.014 | 0 |
| unrestricted best | 79 | 0.376 | 79 |

What the correctness column proves is narrow. Every one of the 150 runs of each set ends final_equiv = equivalent (150 for the calibrated f3 set), which says that the flattened output is formally equal to the input netlist. It does not say the output matches the reference RTL: that netlist itself passes the official testbench in only 128 of the 150 designs (the round-1 self-check; inferred latches and asynchronous resets are the failures), so Loki's output is exactly as correct as its input, no more. The model is checked against the testbench directly, but only its correct samples enter the similarity columns. On operator recovery (RQ-L7) the model is far ahead: on the common Q1 and Q2 designs its operator type-F1 is 0.465 (functionally correct samples elaborated with the same Odin III front as the key) against 0.081 for Loki's calibrated f2 run and 0.074 uncalibrated (the calibrated f3 run recovers no operator, 0.0, so the comparison uses f2, a different floor from the one AST and CFG use). By the rule, H2 is inconclusive: Loki is below the model on both metrics in both quartiles (yes), the equivalence holds in all runs (yes), and the similarity does not stay flat from Q1 to Q4 (no). The positioning this measures is that the model is readable and recovers the operators of small designs but is unverified, and Loki is verified but partial; neither number alone is the result, and the arm that combines them (a model given the ReverseProvenance record, L8) was not run.

RQ-L4: which evidence family drives the posterior? The calibrated flow scores every hypothesis from a per-detector prior plus feature log-likelihood ratios in four families (npn, wl, l2, struct). loki re produces no l2 feature (those exist only in the ReverseProvenance document), so that family is not applicable. For each remaining family we refit the calibration on the train designs of the micro and VerilogEval f2 floors with the family removed and its weight set to 0, and ran the 49 held-out designs (Table XVIII; 1,558 ground-truth instances). Held-out micro-F1 is 0.0159 with the defaults, 0.0038 with the full calibration and 0.0051 with priors only. The full-minus-priors gap is −0.0013, below the 0.02 floor of the rule, so H4 is inconclusive: no family carries measurable performance, because the features add nothing detectable over the per-detector priors. One held-out design (ve_Prob153_gshare) holds most of the ground truth and nothing in it is matched; with it left out (257 instances) the micro-F1 is 0.0553 with the defaults, 0.0225 full and 0.0295 priors only (gap −0.0070), so the verdict does not change; on that subset the defaults score higher than either calibrated row. The fit says so itself: the full calibration sets the npn weight to 0 and shrinks struct and wl (l4_weights.csv). What calibration does is prune, from the many counted constructs of the defaults to a handful, and rank: the AUC of $p_{\text{comp}}$ over the live hypotheses rises from 0.437 to 0.885 (over different pools, since calibration rejects more).

RQ-L5: is the must/may bracket an uncertainty measure? Following the graph-sandwich framing (must is what L1 proved and committed, may adds the candidates it lists), the bracket of a run is read from reverse-provenance.json and its width is the candidate fraction of the cells. H5a: the width correlates negatively with the design's F1. On the uncalibrated VerilogEval f2 runs (90 designs with ground truth) the Spearman correlation is 0.087 ($p = 0.417$), and −0.346 on the calibrated run that lists candidates from $p = 0.1$ (Table XIX), so H5a is not supported. The committed fraction does correlate with F1 (0.485), but trivially: committed cells are the matches. As a soundness check of the bracket itself, 0.840 of the committed cells lie inside the scored ground-truth footprints (must precision), the committed cells cover 0.240 of the truth, and the may bracket contains 0.625 of it. H5b: of 210 cone_check (SAT) calls across all round-2 runs, 166 end equivalent; the largest structural rule that loses no such verdict (adder/out_reads=no) prunes 0.067 of the calls, against the 0.3 the rule asked for (Table XX), so H5b is not supported. The prover budget is spent confirming true hypotheses, and the structural features the calls already carry do not separate the rest; the truth-table checks (2,876) are cheap already. The rule search covers single-bin rules (one feature and one bin of one kind) on the same data, so 0.067 is an upper bound for such rules, not a measured saving; the calls repeat across the eight run sets. The bound that holds for any rule is simpler: a rule that loses no equivalent verdict can prune at most the 44 calls that were not (0.21 of all).

## G. Round 3: fragments or noise, and where the words are

Round 2 left a puzzle. Uncalibrated, the detectors propose many candidates that almost never match (Table XXI); calibrated, they commit almost nothing. Peter's

TABLE XXI
RQ-L9, primary set P (M2N+V2N, f2, uncalibrated): counted candidates, false ones, and the share of the false ones per class (rules of EXPERIMENT.md).

| kind | cand. | false | same ext. | frag. rel. | frag. other | straddle | partial | noise |
|---|---|---|---|---|---|---|---|---|
| ff_en | 1,070 | 1,067 | 0.01 | 0.99 | 0.00 | 0.00 | 0.00 | 0.00 |
| mux_bit | 344 | 336 | 0.00 | 0.61 | 0.39 | 0.00 | 0.00 | 0.00 |
| eq_const | 1,105 | 1,096 | 0.00 | 0.18 | 0.82 | 0.00 | 0.00 | 0.00 |
| register | 155 | 116 | 0.03 | 0.64 | 0.15 | 0.19 | 0.00 | 0.00 |

TABLE XXII
RQ-L9 (H9b), set P: ground-truth instances with at least two cells, the share whose fragment union covers at least 0.8 of them ($q$), the same with candidates of any kind, and the mean fragment coverage.

| kind | GT | matched | $q$ frag. | $q$ any | mean cov. |
|---|---|---|---|---|---|
| ff_en | 156 | 3 | 0.122 | 0.160 | 0.618 |
| mux_bit | 368 | 2 | 0.005 | 0.136 | 0.016 |
| eq_const | 995 | 5 | 0.040 | 0.112 | 0.089 |
| register | 289 | 39 | 0.194 | 0.228 | 0.497 |

question (Appendix A) is why netlists are hard to bring back to HDL when the structure is there to see: muxes, wide signals that behave as words, state feedback with decode and select, and input and output words that mark where the datapath lies. Round 3 is compute-only on the round-2 runs and the L0 ground truth (no language-model call, no new detector). Unlike round 2, its rules were committed before any number was computed (commit 5f8de24, experiments/round3/EXPERIMENT.md); every figure below is a macro generated from the summary CSVs of experiments/round3. The round-2 counts that motivate the question (for instance ff_en 732 candidates and one correct on the held-out designs) were known when the thresholds were set, so H9a was not blind to the data, only to the round-3 numbers. Analyses added after the data was seen are labelled exploratory, are listed in a dated entry of EXPERIMENT.md, and decide nothing. We added the runs the round-2 sets lacked (micro f3, calibrated and not, and micro f1-k6, uncalibrated; one micro f3 design timed out) and score every set, but the decisions use the primary set P: the micro and VerilogEval f2 runs without calibration.

RQ-L9: are the wrong candidates fragments of real words? A false candidate is a counted candidate the one-to-one scorer does not match. We classify each (rules in the protocol): the same extent as a ground-truth instance of its family (a lost tie or a type slip), a fragment of a larger instance of its family, a fragment of an instance of another type, a straddle of several, a partial overlap, or noise. Over the 2,615 false candidates of the four kinds in P, 0.583 (1,525) are fragments of an instance of the matching or a related family. Per kind the related-fragment share is 0.988 for ff_en, 0.638 for register, 0.610 for mux_bit and only 0.175 for eq_const; 3 of four kinds reach 0.5, so H9a is supported by the letter of the rule; not robust. Three qualifications go with it.

TABLE XXIII
RQ-L9 for every run set (four decisive kinds pooled): runs, candidates, false candidates and the share that are related fragments, other-type fragments, noise.

| set | runs | cand. | false | frag. rel. | frag. other | noise |
|---|---|---|---|---|---|---|
| M2N | 65 | 705 | 686 | 0.28 | 0.69 | 0.00 |
| M2C | 65 | 71 | 65 | 0.95 | 0.05 | 0.00 |
| V2N | 150 | 1,969 | 1,929 | 0.69 | 0.30 | 0.00 |
| V2C | not run | | | | | |
| V3N | 150 | 5,005 | 4,997 | 0.29 | 0.71 | 0.00 |
| V3C | 150 | 2 | 0 | 0.00 | 0.00 | 0.00 |
| M3N | 64 | 1,406 | 1,405 | 0.23 | 0.77 | 0.00 |
| M3C | not run | | | | | |
| M1N | 68 | 496 | 469 | 0.38 | 0.57 | 0.00 |

TABLE XXIV
Exploratory, non-deciding additions (2026-10-24, after the data), set P: share of false candidates that are related fragments (by candidate, by distinct cell), the noise′ share (cells only in plain-gate ground-truth words), $q$ over all words and over the words touched by a candidate.

| kind | frag. rel. | frag. rel. (cells) | noise′ | $q$ | $q$ touched |
|---|---|---|---|---|---|
| ff_en | 0.99 | 0.98 | 0.00 | 0.122 | 0.123 |
| mux_bit | 0.61 | 0.61 | 0.01 | 0.005 | 0.006 |
| eq_const | 0.18 | 0.18 | 0.52 | 0.040 | 0.042 |
| register | 0.64 | 0.49 | 0.00 | 0.194 | 0.229 |
| pooled | 0.58 | 0.56 | 0.22 | 0.065 | 0.069 |

First, the noise class is vacuous. The ground-truth instances cover every non-I/O, non-constant cell of the f2 datasets (9,381 of 9,381 in VerilogEval, 9,130 of 9,130 in the micro set), so rule 6 of the protocol could never yield noise, and the noise share 0.000 is structural. The arm of the H9a rule that reads “not supported when the noise share is at least 0.5” was unreachable by construction, and the sentence that no wrong candidate is noise carries no information. As an exploratory replacement we count noise′: a false candidate all of whose cells lie only in plain-gate ground-truth words ($and, $or, $not, $xor, $xnor, the $logic and $reduce gates). Pooled, noise′ is 0.220 (574 candidates); per kind it is 0.522 for eq_const, 0.006 for mux_bit, and 0.000 for ff_en and 0.000 for register (Table XXIV). Half of the false eq_const candidates are thus glue logic, not a piece of a word.

Second, the ff_en fragments are near-tautological. An ff_en candidate has two cells (a flip-flop and its enable LUT), and the ground-truth footprint of a register or $dffe word is the whole word, so a correct per-bit slice of a word counts as a fragment of it; 1,054 of the 1,525 related fragments are ff_en. For ff_en the detector finds correct bit slices and the miss is word-level grouping and extent. Without ff_en (exploratory) the pooled related-fragment share over the other three kinds is 0.304 (1,548 false candidates, 2 of three kinds at or above 0.5), and 0.271 when the largest design is also left out. For mux_bit and register the fragment reading holds at the stated shares; for eq_const it is a detector problem: 0.822 of its false

TABLE XXV
RQ-L10, f2 floor, micro and VerilogEval pooled, relaxed rule ($\tau = 0.75$): the share of ground-truth word constructs seeded by a propagated group within $h$ hops of an I/O port word, and the share never reached.

| kind | class | GT | $h \leq 1$ | $h \leq 2$ | $h \leq 3$ | $h \leq 4$ | never |
|---|---|---|---|---|---|---|---|
| register | datapath | 289 | 0.57 | 0.66 | 0.66 | 0.69 | 0.31 |
| mux | datapath | 90 | 0.40 | 0.41 | 0.41 | 0.41 | 0.59 |
| mux_tree | datapath | 13 | 0.23 | 0.23 | 0.23 | 0.23 | 0.77 |
| adder | datapath | 2,124 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 |
| subtractor | datapath | 20 | 0.20 | 0.20 | 0.20 | 0.20 | 0.80 |
| comparator | datapath | 23 | 0.30 | 0.30 | 0.30 | 0.30 | 0.70 |
| counter | datapath | 23 | 0.00 | 0.00 | 0.00 | 0.00 | 1.00 |
| eq_const | control | 990 | 0.61 | 0.61 | 0.61 | 0.61 | 0.39 |
| fsm | control | 16 | 0.06 | 0.06 | 0.06 | 0.06 | 0.94 |

TABLE XXVI
RQ-L10 by dataset and floor: designs, designs with a port word, input and output port words, and the datapath and control shares reached within two hops.

| dataset | floor | designs | with port word | in words | out words | datapath $h \leq 2$ | control $h \leq 2$ |
|---|---|---|---|---|---|---|---|
| micro | f2 | 68 | 54 | 158 | 22 | 0.35 | 0.68 |
| micro | f3 | 65 | 51 | 150 | 29 | 0.63 | 0.99 |
| micro | f1-k6 | 68 | 54 | 158 | 38 | 0.66 | 0.78 |
| veval | f2 | 150 | 63 | 103 | 13 | 0.07 | 0.60 |
| veval | f3 | 150 | 63 | 103 | 13 | 0.94 | 0.88 |

TABLE XXVII
RQ-L11, ground truth: feedback registers, constant-compare decoders on them, sinks (mux selects and enables) and chains, and the muxes with a select (enabled words) whose select (enable) is a chain sink.

| dataset | floor | designs | with fb. | with chain | fb. reg. | fsm flag | decoders | sinks | chains | mux S chain/all | enable chain/all |
|---|---|---|---|---|---|---|---|---|---|---|---|
| micro | f2 | 68 | 3 | 1 | 3 | 0 | 11 | 3 | 12 | 3/41 | 0/3 |
| micro | f3 | 65 | 3 | 1 | 3 | 0 | 11 | 3 | 12 | 3/41 | 0/3 |
| micro | f1-k6 | 68 | 3 | 1 | 3 | 0 | 11 | 3 | 12 | 3/41 | 0/3 |
| veval | f2 | 150 | 55 | 12 | 197 | 16 | 41 | 30 | 89 | 15/43 | 15/152 |
| veval | f3 | 150 | 55 | 12 | 197 | 16 | 41 | 30 | 89 | 15/43 | 15/152 |

TABLE XXVIII
RQ-L11, detectors: chain elements proposed by a counted candidate with Jaccard at least 0.5 of any kind (in brackets: of the right scored type), chains with all three elements proposed, and ground-truth FSM instances matched by an fsm candidate.

| set | fb. register | decoder | sink | complete chains | fsm |
|---|---|---|---|---|---|
| M2N | 0/3 (0) | 1/11 (0) | 1/3 (0) | 0/12 (0) | 0/0 |
| M2C | 0/3 (0) | 0/11 (0) | 0/3 (0) | 0/12 (0) | 0/0 |
| V2N | 41/197 (26) | 15/41 (9) | 5/30 (0) | 5/89 (0) | 6/16 |
| V2C | 3/197 (0) | 1/41 (0) | 2/30 (0) | 1/89 (0) | 0/16 |
| V3N | 13/197 (3) | 6/41 (2) | 3/30 (0) | 4/89 (0) | 3/16 |
| V3C | 4/197 (0) | 2/41 (0) | 2/30 (0) | 2/89 (0) | 1/16 |
| M3N | 0/3 (0) | 0/11 (0) | 0/3 (0) | 0/12 (0) | 0/0 |
| M3C | 0/3 (0) | 0/11 (0) | 0/3 (0) | 0/12 (0) | 0/0 |
| M1N | 0/3 (0) | 0/11 (0) | 0/3 (0) | 0/12 (0) | 0/0 |

candidates are fragments of an instance of another type (mostly single-LUT $and and $or words, registers and fsm instances), so the LUT computes something that looks like a constant compare and is not one. The mux_bit candidates are 0.390 inside an instance of another type ($shiftx words among them).

Third, the result rests on one design. Most ff_en candidates come from ve_Prob153_gshare, where they sit inside a mux tree, and overlapping candidates inflate their count. With that design left out (exploratory, not the rule) 1,863 false candidates remain, 0.415 of them related fragments and 0.563 fragments of another type, so the pooled share falls under the 0.5 of the rule and H9a would be inconclusive; ff_en stays at 0.966 and register drops to 0.333. Counted over the 4,004 distinct (design, cell) pairs covered by false candidates instead of over candidates (exploratory), the pooled related-fragment share is 0.558. We therefore state H9a with both numbers: supported by the letter of the rule at 0.583, not robust (0.415 without ve_Prob153_gshare, 0.304 without ff_en).

H9b asks whether grouping the fragments would recover the word: the share of ground-truth instances whose fragment union covers 0.8 of them is 0.065 pooled (ff_en 0.122, register 0.194, eq_const 0.040, mux_bit 0.005), and no kind reaches 0.5, so H9b is not supported. The denominator holds words no candidate touched; over the 1,698 words touched by at least one candidate (exploratory) the share is 0.069. The shortfall is therefore detector misses plus non-tiling, not one of them: 0.065 of all words, 0.069 of the touched ones. Even with candidates of any kind the cover stays low (register 0.228, mux_bit 0.136). The fragments exist, but they do not tile the words: a merge step over the present detector output would recover few words. The other sets (Table XXIII) have fewer candidates when calibrated, and the gate floor f3 proposes far more false eq_const candidates than any f2 set.

RQ-L10: do the I/O words find the datapath? A top-level port bus is a word the reverse engineer usually has by name. A group of cells is a propagated word if, bit by bit, a known word drives (or is driven by) one cell per bit of the group with the same cell type, truth table and pin, and at least three quarters of the bits qualify (the strict rule asks for all); we propagate for up to four hops. A ground-truth construct is reached when a group of at most $h$ hops lies for the most part inside it. In the f2 floor, 2,582 datapath word constructs (registers, muxes, adders, comparators, counters) and 1,006 control constructs (eq_const and FSMs) have two or more bits; 0.095 of the datapath and 0.605 of the control constructs are reached within two hops, so H10a is not supported and H10b is not supported. H10b is uninformative by construction once H10a fails: with D = 0.095, "not supported" means $C > D - 0.10 = -0.005$, which holds for any $C \geq 0$. We keep the status the rule gives and do not present it as a finding. Both numbers need reading. The datapath share is pooled and one VerilogEval design alone holds most of the 2,124 adder constructs, which the rule never reaches because the carry chain breaks the bit-for-bit isomorphism; the mean of the per-design shares is 0.326 (datapath, 113 designs) against 0.286 (control, 51 designs; both exploratory), a difference of the sign the hypothesis expects but not of its size. Registers are reached best (0.574 at one hop, 0.664 at two), muxes less, adders and counters almost never, and a second hop adds little: bit-sliced isomorphism stalls after the first layer of logic. The control constructs are reached not because FSM logic

follows the ports but because the constant compares read input ports directly (eq_const 0.614 at two hops); the FSMs themselves are not. The strict rule reaches 0.032 and 0.070, and the gate floor f3 0.914 and 0.894: at f3 a cell group lands inside a gate-level footprint almost by construction, so that figure measures the floor, not the idea. Two details of the counting, checked as exploratory recounts: a $lt, $le, $gt or $ge word with a constant port is counted as a comparator although the protocol says comparators have none (7 such words in f2; excluding them moves the datapath share from 0.0953 to 0.0948, control unchanged); and a $mux inside a mux tree and a register inside a counter are both counted (66 such constructs; deduplicating the footprints gives 0.0938, and both corrections 0.0933). Neither moves the share. The first measure we planned, a construct that is itself a port word, is empty at level 0 by construction (0 port cells in any footprint); read as a group that reproduces the construct's cells at one hop (Jaccard at least 0.5) it is 0.007 for the datapath and 0.047 for the control constructs. The honest summary is that I/O words seed registers and input-side muxes well, and nothing else.

RQ-L11: the FSM chain. Measurement only. In the f2 ground truth of VerilogEval (150 designs) 197 registers feed back to themselves, of which 16 the front end flags as FSM state; 41 constant-compare decoders read such a register, and the chains (register, decoder, select or enable) occur in 12 designs. Of the muxes with a select, 15 of 43 (VerilogEval) and 3 of 41 (micro) take it from such a decoder, 0.214 together, so the FSM, decode, select chain is real but covers a minority of the control muxes of these small designs. The detectors see little of it (Table XXVIII): in V2N they propose 41/197 feedback registers, 15/41 decoders, 5/30 sinks, all three elements of 5/89 chains, and 6/16 FSM instances as an fsm candidate. The chains are triples (register, decoder, sink) and reuse their elements: the 89 triples hold only 30 distinct sinks and 41 distinct decoders. Counted on distinct elements, the detectors propose 5/30 of the sinks and 15/41 of the decoders, against 5/89 complete triples.

Implication for a word-first detector. The classification cannot say that the wrong candidates are not noise, because the ground truth covers every cell and the noise class was unreachable; the exploratory noise′ share is 0.220 pooled and 0.522 for eq_const. The fragment share depends on the kind and, for the pooled figure, on one design and on the ff_en slices: ff_en, mux_bit and (less so) register fail as pieces of larger constructs, while eq_const fails as a detector problem (the local pattern is too weak, and a word-level view, the select and enable fan-out across a word, must decide whether a compare is a decoder). Because the fragments do not tile the words, grouping the present candidates is not enough, and a detector should construct the word first and score it as a unit. Concretely, we propose to find (i) register words by shared clock, enable and reset across flip-flops; (ii) datapath words by per-bit cones that are NPN-isomorphic, seeded from the ports but not limited to them (the port words reach registers and nothing past the first layer); (iii) control nets by select and enable fan-out across a word; and (iv) FSMs by a feedback strongly connected component plus the decoders on it. Related work attacks the word problem along the same lines: WordRev [6] derives word-level structures from the gate-level netlist of a circuit and lets the user specify sequences of word-level operations whose gates it then extracts, and DANA [7] recovers high-level registers by analysing the flow of data between individual flip-flops. We have not run either tool on these sets and do not claim a comparison. The next round builds this detector (L8, the language model given the ReverseProvenance record, stays deferred).

TABLE XXIX
The round-4 corpus: designs per family and split, and designs that build on f1-k6, f2 and f3 with every equivalence check passing.

| family | train | iid test | ood test | ok on all floors |
|---|---|---|---|---|
| adder | 70 | 10 | 20 | 100 |
| alu | 70 | 10 | 20 | 100 |
| comparator | 70 | 10 | 20 | 100 |
| counter | 70 | 10 | 20 | 100 |
| dpfsm | 70 | 10 | 20 | 100 |
| fsm | 70 | 10 | 20 | 100 |
| mux | 70 | 10 | 20 | 100 |
| pipeline | 70 | 10 | 20 | 100 |
| regfile | 70 | 10 | 20 | 100 |
| register | 70 | 10 | 20 | 100 |
| shifter | 70 | 10 | 20 | 100 |
| subtractor | 70 | 10 | 20 | 100 |

TABLE XXX
H12b on real designs (micros and the 33 VerilogEval test designs, f2): pooled F1 of the dictionary (V0 as written; V1 and V2 added after Stage 0) and of the better detector arm, and the number of designs and ground-truth constructs.

| kind | dict V0 | dict V1 | dict V2 | best detector | designs | GT |
|---|---|---|---|---|---|---|
| adder | 0.000 | 0.012 | 0.008 | 0.208 | 29 | 69 |
| subtractor | 0.000 | 0.000 | 0.000 | 0.129 | 14 | 29 |
| comparator | 0.000 | 0.002 | 0.010 | 0.003 | 27 | 663 |
| eq_const | 0.000 | 0.000 | 0.008 | 0.001 | 25 | 643 |
| mux | 0.027 | 0.031 | 0.013 | 0.025 | 22 | 71 |
| register | 0.000 | 0.026 | 0.080 | 0.104 | 41 | 240 |
| ff_srst | 0.000 | 0.019 | 0.021 | 0.000 | 12 | 15 |

## H. Round 4: patterns from our own data, big designs, and what is recoverable

00.000 10.158 20.054

Round 3 pointed to a word-first detector. Round 4 builds what such a detector needs: labelled data of our own, a dictionary of netlist patterns mined from it, a look at what the larger VTR designs add, and a statement of what the flow destroys. Peter (Appendix A) asked for "patterns from our data (what about the bigger designs, do they provide other insights of patterns)". No language-model call and no new detector in loki/ were used; the pro-

TABLE XXXI
GNN macro-F1 (f2, mean of 3 seeds) by corpus train size, on four test sets; last column: trained on the VerilogEval train split.

| test set | 36 | 72 | 144 | 288 | 576 | 840 | VE train |
|---|---|---|---|---|---|---|---|
| iid | 0.336 | 0.358 | 0.367 | 0.374 | 0.402 | 0.423 | 0.084 |
| ood | 0.244 | 0.308 | 0.294 | 0.306 | 0.325 | 0.346 | 0.098 |
| micro | 0.147 | 0.105 | 0.126 | 0.115 | 0.111 | 0.122 | 0.068 |
| vetest | 0.042 | 0.036 | 0.043 | 0.031 | 0.030 | 0.031 | 0.169 |

TABLE XXXII
Slope of macro-F1 per doubling of the corpus train size, with 95% bootstrap interval over test designs.

| floor | test set | designs | slope | 95% CI |
|---|---|---|---|---|
| f2 | iid | 118 | 0.0171 | [0.0053, 0.0261] |
| f2 | ood | 231 | 0.0174 | [0.0092, 0.0234] |
| f2 | micro | 62 | −0.0039 | [−0.0100, 0.0031] |
| f2 | vetest | 29 | −0.0026 | [−0.0097, 0.0094] |
| f3 | iid | 118 | 0.0077 | [0.0005, 0.0151] |
| f3 | ood | 231 | 0.0111 | [0.0067, 0.0146] |
| f3 | micro | 59 | −0.0022 | [−0.0048, 0.0006] |
| f3 | vetest | 29 | −0.0038 | [−0.0071, −0.0009] |

totypes live in experiments/round4. The rules were committed before the corpus was generated (commit c6c66bf, experiments/round4/EXPERIMENT.md); two variants of the dictionary (V1, V2) were added in a dated entry after Stage 0 and before any dictionary result, and they are reported next to the rule as written (V0), which alone gives the primary status. The thresholds were set knowing the round-2 and round-3 counts and the ground-truth kind counts of the real designs (which fixed the seven eligible kinds of H12b).

RQ-L12a: a synthetic corpus. A seeded generator of twelve idiom families (muxes, counters, adders, subtractors, comparators, shifters, FSMs, register files, pipelines, registers, a datapath with an FSM controller, an ALU), each parameterised by width, depth, encoding, reset and enable style and coding style, gives 1,200 designs split by parameter region: train values only (840), the same region with unseen text (120, iid), and at least one test-only value, such as an unseen width or coding style (240, ood). Built by Odin III on f1-k6, f2 and f3, 1,200 of 1,200 pass every equivalence check on all three floors (Table XXIX); H12a is supported. Forty-one corpus designs have no labelled logic cell (constant shifts, constant-only comparators) and are excluded from the GNN train and test sets (the trainer needs a labelled logic node per train design); the dictionary scores them with no candidates and no ground truth.

RQ-L12b: a signature dictionary. For each cell the signature is the canonical class of the Boolean function of its largest cut and the Weisfeiler-Lehman context of radius $r$; the dictionary keeps, per construct kind, the signatures that occur at least three times in the corpus train split and lie in that kind at least 60% of the time;

TABLE XXXIII
Big VTR designs at f3: cells, Odin III peak RSS (MB, all three floors), words with at least two cells and width $\geq 2$, share of them in a repeat group, memories, buses of at least 8 bits, nets with at least 8 readers, FSM words.

| design | cells | RSS | words | repeated | mem | buses | hi-fanout | fsm |
|---|---|---|---|---|---|---|---|---|
| blob_merge | 31,778 | 4915.6 | 1,090 | 0.744 | 0 | 51 | 717 | 1 |
| sha | 10,629 | 841.7 | 225 | 0.924 | 0 | 29 | 99 | 0 |
| stereovision0 | 56,208 | 1665.9 | 1,814 | 0.942 | 0 | 1263 | 182 | 0 |
| stereovision1 | 142,932 | 1863.8 | 1,892 | 0.972 | 0 | 830 | 1430 | 48 |
| stereovision2 | 254,419 | 4793.1 | 2,001 | 0.982 | 0 | 951 | 1724 | 0 |
| stereovision3 | 859 | 75.8 | 145 | 0.834 | 0 | 9 | 10 | 1 |

TABLE XXXIV
Style-recovery ceiling (classes of equivalent netlists over coding styles) on the twin set, per family and floor.

| family | f1-k6 | f2 | f3 |
|---|---|---|---|
| all | 0.470 | 0.482 | 0.500 |
| adder | 0.667 | 0.667 | 0.667 |
| alu | 0.333 | 0.472 | 0.528 |
| comparator | 0.333 | 0.333 | 0.333 |
| counter | 0.333 | 0.333 | 0.333 |
| dpfsm | 0.500 | 0.500 | 0.500 |
| fsm | 0.500 | 0.500 | 0.500 |
| mux | 0.562 | 0.562 | 0.667 |
| pipeline | 0.333 | 0.333 | 0.333 |
| regfile | 0.500 | 0.500 | 0.500 |
| register | 0.333 | 0.333 | 0.333 |
| shifter | 0.625 | 0.625 | 0.625 |
| subtractor | 0.667 | 0.667 | 0.667 |

a matcher labels the test cells and joins labelled cells into words by adjacency or by a shared net. Against the round-2 detectors on the micros and the 33 held-out VerilogEval designs (f2, seven eligible kinds) the dictionary as written (V0) wins on none: H12b is not supported (won=0/7 floor=f2). V1 (cuts of up to six leaves) and V2 (a shared clock also joins storage cells), added after Stage 0, also win on none (won=0/7 floor=f2; won=0/7 floor=f2) and change no status. Table XXX gives the F1 per kind: on the real designs the dictionary finds almost nothing the corpus signatures were mined for. It does not even generalise inside the corpus: the iid test micro-F1 is low and H12b2 is inconclusive (R=0.3288 iid=0.1866 ood=0.0614 (iid $<$ 0.20: ratio uninformative); V2: R=0.6429 iid=0.2576 ood=0.1656). Signatures by cone class alone are mostly floor-specific (l12b_stability.csv): few classes that fire for a kind at f2 also fire at f3.

RQ-L12c: GNN scaling. Trained on corpus subsets of 36 to 840 designs (three seeds each), the GNN improves on

TABLE XXXV
MDL readability: description length of a candidate relative to the VerilogEval reference (median ratio) and the share of designs where the reference is strictly shorter.

| variant | designs | median ratio | ref shorter |
|---|---|---|---|
| ref | 156 | 1.00 | 0.000 |
| llm | 73 | 1.00 | 0.260 |
| l0-f2 | 150 | 3.63 | 0.860 |
| l0-f3 | 150 | 3.31 | 0.913 |
| f2-none | 150 | 7.89 | 0.940 |
| f3-none | 150 | 15.40 | 0.940 |
| f2-calib | 150 | 4.24 | 0.867 |
| f3-calib | 150 | 3.31 | 0.913 |

the corpus test sets but not on real designs (Tables XXXI, XXXII): on the 29 VerilogEval held-out designs with a labelled logic cell the slope is −0.0026 per doubling (95% interval [−0.0097, 0.0094]), so H12c is not supported. A GNN trained on the VerilogEval train split itself reaches a higher macro-F1 on its held-out designs than any corpus size.

RQ-L13: big designs. Table XXXIII lists the VTR designs built one at a time with Odin III's peak resident set measured. Big designs are not equivalence-checked (agent default, as in Odin III itself; meta.json records skipped:big). 13 designs built. Odin III's reader refuses the f3 floor (unresolved_module) of mkPktMerge, spree, boundtop, raygentop, or1200, mkSMAdapter4B, mkDelayWorker32B, whose f1-k6 and f2 floors are built (the mkPktMerge case among them); bgm, LU8PEEng, LU32PEEng exceeded the 14 GiB address limit (60% of the VM) and are skipped, partly or wholly. arm_core had not finished after more than three CPU hours (a hang in the linked ABC is documented for this design in the Odin III notes) when this section was written, and mcml and LU64PEEng (which Odin III's goldens do not cover) were not started; the results use the designs listed in Table XXXIII. Share of words in a repeat group in the small designs: 0.688 (1,055 design-floors). H13a is supported (share_rep(big)=0.9283 (6653/7167) designs=6), H13b supported (big=0.9283 small=0.6879 (6233/9061) diff=0.2404) and H13c supported (same_signature=0.9374 groups=911).

RQ-L14: what is recoverable. On the twin set (the same design in every coding style) the style-recovery ceiling is 0.470 at f1-k6, 0.482 at f2 and 0.500 at f3 (Table XXXIV): H14a is supported and H14b inconclusive (f3-f2=0.0177). The MDL score ranks the reference above the recovered RTL in the shares of Table XXXV (H14c inconclusive); its correlation with the round-2 AST similarity is rho=-0.5110 (p=1.1e-35) partial=0.3029 (p=2.1e-12) n=516 (H14d inconclusive), and the partial correlation controlling for size does not support a readability reading: the score is largely a size measure.

What it means for the round-5 word-first detector. The data say what not to do: a signature dictionary mined from idiom-shaped synthetic designs does not transfer to real logic (H12b, H12c), so the corpus is a test bed for word-grouping rules, not a source of a lookup table. The detector should build words from structure that is stable across designs: registers by shared clock, enable and reset; datapath words by per-bit isomorphic cones; control nets by select and enable fan-out; FSMs by feedback components and their decoders, with the eq_const decoder decided at word level. The repeat measures of Table XXXIII say how far repetition gives word evidence for free.

## VI. Discussion

Three observations follow. First, the answer depends on what the flow erased. The structures that survived synthesis recover almost perfectly without any intelligence beyond reading the pass's own facts; the structures synthesis erased (the counter behind its flip-flop, the mux behind its gates) recover weakly, and no amount of honest Bayesian machinery changes that without better detectors or a model. Second, the graph features carry the recovery: the GNN only beats 1-WL when node, graphlet, WL and NPN features combine, which is the same lesson as the clustering table (community methods find the hierarchy, centroid methods do not). Third, the benchmarks made the limits visible: the renamed-instance protocol proved naming neutrality over 105,646 identifiers, and the same protocol exposed a rename-script defect (library .blackbox model names were renamed, which silently removed every hard proposal on renamed floors) and the spectral determinism defect – both fixed or reported, and both invisible without the protocol.

A fourth observation comes from round 2 (Section V-F): readable and verified are different properties, and the two tools hold one each. The language model produces RTL whose similarity to the reference is high and whose operators it names (operator type-F1 0.465 on Q1 and Q2, correct samples elaborated to RTL), but nothing proves the text equals the netlist; Loki's output is formally equal to its input in all 150 runs and recovers few operators (0.081, calibrated f2 run; the f3 run recovers none), and its similarity to the reference is lower in every cell. The f3 failure of round 1 may be the detectors' vocabulary and not missing information (a LUT6 re-map reaches the f2 level in-sample but not on the held-out designs, 0.87 against 0.51, so the question stays open), the three refinements we expected to help (evidence-family weighting, the width of the must/may bracket as an uncertainty measure, a structural prefilter before the prover) did not on the registered tests, and the one new instrument that earned its place is the operator F1, which sees recovery where AST and CFG see only form.

A fifth observation comes from round 3 (Section V-G): many wrong candidates are pieces of real constructs, but

the pieces do not tile the words. By the letter of the pre-registered rule about half of the false candidates are fragments of an instance of a related kind (H9a supported by the letter of the rule; not robust: 0.415 without one design, 0.304 without the near-tautological ff_en slices), and the noise class could not be filled because the ground truth covers every cell. A word's fragments rarely cover it (H9b not supported). Propagation from the I/O words reaches registers but not the arithmetic or the FSMs (H10a not supported; H10b is not supported but uninformative once H10a fails), and the FSM, decode, select chain covers 0.214 of the control muxes. The next detector is therefore word-first, not a regrouping of the present one.

A sixth observation comes from round 4 (Section V-H): patterns mined from synthetic idioms do not transfer to real logic, and the flow destroys part of the style. The dictionary wins on no eligible kind (H12b not supported) and the GNN does not improve on real designs with more synthetic data (H12c not supported); on the twin set the style-recovery ceiling is 0.500 at f3.

## VII. Conclusion

Loki III asks how far netlist-to-RTL reverse engineering gets without a language model. On the benchmarks of this paper the answer is: the already-reverse-engineered structures, perfectly; the rest, weakly but measurably, with calibration that reports its own error, a graph stage whose features beat 1-WL, and clusters that carry the designer's hierarchy. The methodology – provenance-tracked labelled datasets plus the renamed-instance protocol – is the part we would export to any future comparison, with or without a model.

## Appendix A
## Research log

This appendix records the order in which the questions were asked, what each result suggested, and what became of the next set of hypotheses, so that the progress of the work is visible and not only its end state. The decision rules of round 2 are in experiments/round2/EXPERIMENT.md; their thresholds were set by the implementer during the round (the file was committed after the scripts and the result CSVs), so they are not pre-registered. The rules of round 3 (experiments/round3/EXPERIMENT.md) were committed before any round-3 number was computed (commit 5f8de24).

Round 0 (2026-10-21 to 22): build. Question: how much structure does Loki recover without a language model? Built L0 (provenance-labelled, anonymised datasets and the renamed-instance neutrality check), L1 (Bayesian hypotheses with calibration and proofs) and L2 (graph analysis).

Round 1 (2026-10-23): first results. Section V-B to Section V-E. Hard blocks are fully recovered, because synthesis had already reverse engineered them; soft structures recover weakly; on the cmos6 gate netlist (f3) L1 recovers nothing and AST and CFG do not move; the GNN beats 1-WL; and the language model is strong on the simple VerilogEval quartiles but its output is unverified.

What round 1 suggested. (i) The f3 zero beside a weak f2 result: is the failure the detectors' LUT vocabulary, not missing information? (ii) The language-model and Loki figures sat on different benchmarks: put them on the same 150 designs. Our working position is that the model is readable but unverified on small designs, Loki is verified but partial, and the combination is the contribution. (iii) The calibrated flow commits little: which evidence family earns that? (iv) The reverse-engineering result is a smaller graph bracketed by what is proven (must) and what is possible (may), in the style of the graph-sandwich problem: is the width of that bracket an uncertainty measure, and does a cheap structural test before the prover keep recall? (v) AST and CFG tie where the recovery differs: is there a metric that sees it?

Round 2 (2026-10-24): the hypotheses and what happened. Table XXXVI. The operator F1 is the one hypothesis the rule supports, by its letter and on weak evidence (H7). The LUT6 re-map reaches the f2 level in-sample (0.87) but not on the held-out split (0.51, too few matches for the rule), so H1 is inconclusive. The head-to-head puts Loki below the model on every cell with its output equal to its input in all runs, but the similarity does not stay flat with complexity (H2, inconclusive). The evidence families add nothing over the priors on held-out designs, because the calibration already zeroes or shrinks them (H4, inconclusive). The bracket width is not an uncertainty measure on the registered set (H5a, not supported; an exploratory look at calibrated runs that list candidates from $p = 0.1$ gives Spearman $-0.346$), and a structural prefilter cannot cut the prover without losing proved constructs (H5b, not supported). The head-to-head (Section V-F) fills in the positioning of the discussion.

What round 2 suggested. The calibrated flow commits almost nothing and the uncalibrated one proposes many wrong candidates. Are the wrong candidates fragments of real word constructs (a grouping problem) or unrelated (a detector problem)? Peter (2026-10-24) put the same question from the reverse-engineering side: "for Loki - the question is from a reverse engineering perspective why is it hard to reconstruct to HDL - going from if structures (which we know are muxes) and then down it seems like many of the patterns are constant - so hypothesizing that things are a bus (and by a bus I mean a signal that could be a traditional bus, but also a wide wordsize signal that has regular oprtations to it - for example figuing out that there is a word in the datapath or control then allows us to look for patterns to reconstruct. Control flow is usually in the form of muxin and FSMs have pattern of feedback (state equals next state) and then control based on state. How do we find these patterns better?" and "Also, the outputs (words) and the inputs as words also provide hints

TABLE XXXVI
Research log: the round-2 questions, their status and the deciding number (all generated from experiments/round2).

| RQ | Hypothesis | Round | Status | Deciding number |
|---|---|---|---|---|
| L1 | f3 failure is detector vocabulary | 2 | inconclusive | micro-F1 LUT6/f2 = 0.87 (all, in-sample), 0.51 (test) |
| L2 | Loki below the LLM on AST/CFG, flat over Q1–Q4 | 2 | inconclusive | AST Q1/Q2 0.763/0.460 vs 0.894/0.761 |
| L4 | one or two evidence families carry the posterior | 2 | inconclusive | full minus priors-only micro-F1 −0.0013 |
| L5a | bracket width falls with per-design F1 | 2 | not supported | Spearman 0.087 ($p$ = 0.417) |
| L5b | structural prefilter cuts prover calls, keeps recall | 2 | not supported | best zero-loss rule prunes 0.067 |
| L7 | operator F1 separates AST/CFG ties | 2 | supported by the letter of the rule (weak evidence) | tie pair dF1 0.0173; flatten dAST 0.0422 |
| L3 | cross-suite GNN | next | not run | – |
| L6 | motif null model | next | not run | – |
| L8 | LLM with reverse provenance | next | deferred (Peter) | – |

TABLE XXXVII
Research log: the round-3 questions, their status and the deciding number (all generated from experiments/round3).

| RQ | Hypothesis | Round | Status | Deciding number |
|---|---|---|---|---|
| L9a | wrong candidates are fragments of GT words | 3 | supported by the letter of the rule; not robust | related-fragment share 0.583 of 2,615 false; noise 0 by construction (every cell is in a GT word); without the largest design 0.415, without ff_en 0.304 (exploratory) |
| L9b | fragments of a word cover it | 3 | not supported | share with coverage $\geq$ 0.8: 0.065 (touched words only, exploratory: 0.069) |
| L10a | datapath words reached from I/O within 2 hops | 3 | not supported | share reached 0.095 (per-design mean 0.326, exploratory) |
| L10b | control words are not | 3 | not supported (uninformative by construction) | share reached 0.605 (per-design mean 0.286, exploratory) |
| L11 | FSM chain: feedback, decode, select | 3 | measurement only | 0.214 of muxes with a select are chain sinks |

TABLE XXXVIII
Research log: the round-4 questions, their status under the Stage-0 rules and the deciding number (generated from experiments/round4).

| RQ | Hypothesis | Round | Status | Deciding number |
|---|---|---|---|---|
| L12a | corpus is usable | 4 | supported | P_ok=1.0000 generated=1200 ok=1200 min_family_ok=100 ood=1.0000 train=1.0000 |
| L12b | dictionary beats detectors (V0 as written) | 4 | not supported | won=0/7 floor=f2 |
| L12b | same, V1 (added after Stage 0) | 4 | not supported | won=0/7 floor=f2 |
| L12b | same, V2 (added after Stage 0) | 4 | not supported | won=0/7 floor=f2 |
| L12b2 | dictionary generalises over unseen regions (V0) | 4 | inconclusive | R=0.3288 iid=0.1866 ood=0.0614 (iid < 0.20: ratio uninformative) |
| L12c | GNN transfer improves with corpus size | 4 | not supported | slope=-0.0026/doubling CI=[-0.0097,0.0094] n_designs=29 |
| L13a | big designs mostly repeated structure | 4 | supported | share_rep(big)=0.9283 (6653/7167) designs=6 |
| L13b | big designs repeat more than small | 4 | supported | big=0.9283 small=0.6879 (6233/9061) diff=0.2404 |
| L13c | repeated instances share the signature | 4 | supported | same_signature=0.9374 groups=911 |
| L14a | flow destroys style information | 4 | supported | ceiling_f3=0.5000 f2=0.4823 f1=0.4697 |
| L14b | coarser floor destroys more | 4 | inconclusive | f3-f2=0.0177 |
| L14c | MDL ranks references above recovered RTL | 4 | inconclusive | l0-f2=0.860 l0-f3=0.913 f2-none=0.940 f3-none=0.940 f2-calib=0.867 f3-calib=0.913 |
| L14d | MDL tracks round-2 similarity | 4 | inconclusive | rho=-0.5110 (p=1.1e-35) partial=0.3029 (p=2.1e-12) n=516 |

of where the words are".

Round 3 (2026-10-24): the hypotheses and what happened. Table XXXVII; Section V-G. Rules committed first (5f8de24). H9a (wrong candidates are related fragments) is supported by the letter of the rule, with related-fragment share 0.583 of 2,615 false candidates, but not robust: 0.415 without the largest design and 0.304 without ff_en (exploratory), and the noise class is unreachable by construction (the ground truth covers every cell); H9b (fragments cover the word) is not supported at 0.065; H10a (datapath words reached from the ports within two hops) is not supported at 0.095 and H10b (control is not) not supported at 0.605, uninformative by construction once H10a fails; the FSM chain (L11) is measured only. The next step is a word-first detector (register words by shared clock, enable and reset; datapath words by per-bit isomorphic cones; control nets by select fan-out; FSMs by feedback components and their decoders); L8 stays deferred.

Next. L3, a cross-suite GNN (train on one suite, test on the other); L6, a motif null model for the graphlet counts; and L8, a language model given the ReverseProvenance document (deferred by Peter), which is the experiment that tests the combination directly.

Round 4 (2026-10-25): the hypotheses and what happened. Table XXXVIII; Section V-H. Peter (2026-10-24): "Queue those as round 4 after round 3 finishes - we're trying to make these better - I like the Loki ideas so yes (especially the idea of patterns from our data (what about the bigger designs, do they provide other insights of patterns)." What round 3 suggested: the wrong candidates are real constructs that do not tile the words, so a word-first detector needs labelled data and pattern knowledge. Rules committed first (c6c66bf); variants V1 and V2 of the dictionary were added after Stage 0 and before any dictionary result (entry 2026-10-25-a) and change no status. RQs: L12a corpus (H12a supported), L12b dictionary (H12b not supported, H12b2 inconclusive), L12c GNN scaling (H12c not supported), L13 big designs (H13a–c supported, supported, supported), L14 recoverability (H14a–d supported, inconclusive, inconclusive, inconclusive). Next: the word-first detector of the discussion, built and tested on the corpus.

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